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SA 46. SMOTRE NAUČNIH RADOVA STUDENATA POLJOPRIVREDE I
VETERINARSKE MEDICINE SA MEĐUNARODNIM UČEŠĆEM**

**PROCEEDINGS
OF THE 46th CONFERENCE FOR STUDENTS OF AGRICULTURE AND
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Council of the University of Novi Sad, Faculty of Agriculture, at its fourth meeting on 18th of February 1981 made a decision that the Conference for students of agriculture and veterinary medicine with international participation, held each year at the Faculty of Agriculture will be held in the memory of academician ***Dr Petar Drezgić***, professor, and the prizes awarded at the Conference will bear his name.

PREFACE

The first Conference for students of agriculture and veterinary medicine with international participation was held in Novi Sad 46 years ago. To the present date, many papers from national and international authors were presented (Hungary – Gödölö and Debrecen, Germany – Kassel, Slovakia – Nitra, Poland – Wroclaw and Warsaw, Romania – Timisoara, Cluj and Bucharest, Bulgaria – Plovdiv, Russia – Moscow and Kemerovo, Macedonia – Skopje and Bitola, Bosnia and Herzegovina – Sarajevo, Banja Luka, Croatia – Zagreb). Since 2009, papers presented at the Conference are published in a special publication named “PROCEEDINGS OF THE 46th CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY MEDICINE WITH INTERNATIONAL PARTICIPATION” where all published papers have all elements of the original scientific paper. Starting this year, we enabled students and mentors who are interested in benefitting more from this Conference to publish their papers in the journal Contemporary Agriculture published by the University of Novi Sad, Faculty of Agriculture.

Also, besides the Faculty of Agriculture from Novi Sad, co-organizer of the Conference is also Faculty of Agriculture from Timisoara and there is a prospect of joining another University center from the region. This Conference is strongly supported by the Student Parliament of the Faculty of Agriculture, Novi Sad.

As the host and the principal organizer of the Conference University of Novi Sad, Faculty of Agriculture has a special honor and a pleasure to wish a warm welcome to all participants, successful work and amusing socializing in Novi Sad.

Vice-dean for science and international cooperation
Dr Branko Ćupina, full professor

President of the organizing committee
Dr Dragana Budakov, associate professor

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INFLUENCE OF DIFFERENT REDUCING AGENTS ON GRAŠEVINA WHITE WINE AROMATIC PROFILE

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SUMMARY

Graševina is the most planted grape variety in Republic of Croatia with great quality potential. The quality of grapes, and thus of wine, largely depends on the conditions in which they are grown. If they are grown in appropriate conditions, they can give high quality wines. However, like all white wines, Graševina wine needs protection from oxidation and discoloration, for which sulfur dioxide (SO₂) is most often used. As the most commonly used preservative in wine due to the antioxidant, antimicrobial and coagulation properties, its use is almost indispensable, although it can cause allergic reactions in a part of sensitive population. Therefore, consumers are increasingly reaching for wines produced with alternative procedures or agents that prevent oxidation and discoloration, such as glutathione. In this study, the quality and chemical composition of Graševina wines, produced with SO₂ and alternative agents, were compared. The analysis showed significantly higher concentrations of total fatty acids, esters and higher alcohols in Graševina wine produced with SO₂, together with more quality sensory properties.

Key words: sulfur dioxide, glutathione, antioxidants, inactive yeast

INTRODUCTION

Graševina is a grape variety (*Vitis vinifera* L.) whose origin has not been determined yet. Enologically, it is an extremely valuable variety because it is suitable for obtaining different wine styles and categories, which is one of the main reasons for being the most common grape variety in Croatia. The Graševina wines are characterized by a golden-yellow, yellow or yellow-green color, floral to fruity aroma and usually are in category of still dry wines, and more rarely like predicate or sparkling wine. The quality varies significantly, depending on the environmental conditions of the location, vintage and yield.

Sulfur dioxide (SO₂) is the most commonly used enological preparation for preservation of wine. The use of SO₂ in winemaking was officially allowed in the 17th century, which makes it the oldest wine preservative. SO₂ has unique triple property, antioxidative, antimicrobial and coagulative. It binds to phenolic compounds and prevents their oxidation (Martinec, 2012). Oxidation is an unwanted process in wine that results in negative aromas and brown color of wine. Sulfur dioxide protects aromatic compound groups

in wine like higher alcohols, esters, fatty acids, terpenes and C-13 norisoprenoids. Higher alcohols are volatile compounds, formed as a secondary product of yeast metabolism, that are important precursors for the formation of esters and aldehydes. Esters are created by the reaction of an organic acid with alcohol, and they affect the sensory properties of wine, by fruity aromas. C13-norisoprenoids are formed by oxidative degradation of carotenoids and significantly contribute to the aroma due to the low detection level. Fatty acids have a primary role as precursors of other aromatic compounds. Terpenes can be free (volatile) and bound to sugars (non-volatile). Free terpenes enrich the aromatic profile of wine, while bound terpenes make the aromatic potential of wine that can be released during the wine production and ageing. Despite all good sides, in certain concentrations, sulfur dioxide can be harmful to wine consumers, so it should be handled according to recommendations and production conditions (Stockley *et al.*, 2021).

Due to the negative correlation between SO₂ and health, there is an increasing demand for wines with less sulfur dioxide, so-called "natural wines". However, the production of wine without sulfur dioxide faces many problems. Such wines are more prone to premature aging, variations in the bottles, the appearance of unwanted aromas, high volatile acidity and the appearance of *Brettanomyces* yeasts associated with wine spoilage. In order to avoid this negative consequences, wine producers started to use natural, alternative preparations. One of them is glutathione, a tripeptide consisting of L-glutamate, L-cysteine and glycine, naturally presented in grapes mostly in reduced form (GSH) (Kritzinger *et al.*, 2012).

MATERIAL AND METHODS

The research was carried out during the harvest 2022, at the experimental viticultural and enology station Jazbina at Faculty of Agriculture. The grapes were picked by hand and processed quickly. After harvesting, the primary processing was carried out, and during further processing of the must, two treatments were carried out: with SO₂ and with alternative protocol without SO₂ (glutathione, bioprotective inactive yeast, antibacterial biopolymer). Alternative protocol implied the treatment of grapes with inactive yeast *Metschnikowia pulcherrima* (LEVEL2 INITIA™, Lallemmand) after the harvest and cooling down to 8 °C for 24 h. During the pressing, GLUTASTAR™, Lallemmand (0.2 g kg⁻¹) was added to the vat under the press. The grapes were pressed with a hydraulic press at a pressure of 2 bar. Pectolytic enzyme Lallzyme CUVÉE BLANC™, Lallemmand (2 g hL⁻¹) was added to the must together with Bactiless™, Lallemmand (29 g hL⁻¹), a natural biopolymer obtained from the fungus *Aspergillus niger*. For treatment with SO₂, 5 %-sulfuric acid (SUMPOvin) was added to the vat under the press and to the must before settling (1 mL L⁻¹). The must was settling down for 24 h in the refrigeration chamber at 10 °C.

After clarifying and decanting from the lees, the must from each treatment was poured into 25 L-demijohns in duplicate. The alcoholic fermentation was carried out under the same conditions in both treatments. This was followed by inoculation with the selected *Saccharomyces cerevisiae* yeast (Lalvin™ EC-1118, Lallemmand) and rehydration of the yeast by addition of yeast starter (Goferm Protect Evolution™, Lallemmand, 30 g hL⁻¹). After one-

third of fermentation, the yeast was fed by complex nutrient Fermaid E™, Lallemand (20-30 g hL⁻¹) to prevent sluggish or stuck fermentation.

At the end of alcoholic fermentation, Reduless™, Lallemand (1 g hL⁻¹) yeast derived product with biologically bound copper developed to reduce sulfur compound off-flavors and improve wine quality was added to Graševina wine no-SO₂ treatment, 24 h before decanting. After decanting, Noblesse™ (Lallemand), partially autolyzed yeast derivative which contributes to balance, sweetness and softness on the finish, was added to young wine. SUMPOvin (1 mL L⁻¹) was added to Graševina yeast with SO₂ after the alcoholic fermentation finished. Both wines were stored under cellar conditions until the physicochemical and sensory analysis.

During the alcoholic fermentation, sugar content was measured every two days using a digital and analog refractometer (Zoecklein *et al.* 1995). Sugar concentration was expressed in degrees Oechsle (°Oe). After three months, physicochemical analysis was carried out according to standard methods (O.I.V., 2019). The parameters analyzed were specific gravity, alcohol, total extract, reducing sugars, sugar-free extract, total acidity as tartaric acid, volatile acidity as acetic acid, pH, free SO₂, bound SO₂, total SO₂ and ash. Analysis of volatile compounds was conducted on TRACETM 1300 Gas Chromatographer coupled to ISQ 7000 TriPlus quadrupole mass spectrometer (Thermo Fisher Scientific Inc., USA) equipped with TG-WAXMS A capillary column (60 m×0.25 mm×0.25µm film thick-ness; Thermo Fisher Scientific, USA).

Sensory evaluation was performed by six qualified evaluators by Quantitative Descriptive Analysis. The sensory properties evaluated were color, odor, taste, body, harmony and overall impression. Each property was given a score from 0 to 5, depending on the intensity of a particular characteristic (Herjavec *et al.*, 2007).

Statistical analysis was performed by using the Analysis of Variance (ANOVA). In order to compare variable means and to examine which wines were different, Tukey's HSD test was used as a comparison test when samples were significantly different after ANOVA ($p < 0.05$) of analyzed compounds.

RESULTS AND DISCUSSION

Basic grapes and wine analysis

The basic parameters of Graševina grape must were 114 °Oe, 8.7 g/L of total acidity and pH value 3.05. It could be concluded that potential alcoholic strength and high acidity will give unbalanced wine.

In table 1. the basic physiochemical composition of these wines can be seen. The specific gravity is within the permissible limits and the difference between two treatments was inconspicuous. The alcoholic strength was high for this grape variety in this winegrowing hill. The total extract met the requirement for high quality wines. The wines were classified as dry wines due to the residual sugar ($< 4 \text{ g L}^{-1}$). The total acidity was high for still white wines and it did not follow the high alcohol content. The pH was low as expected due to the

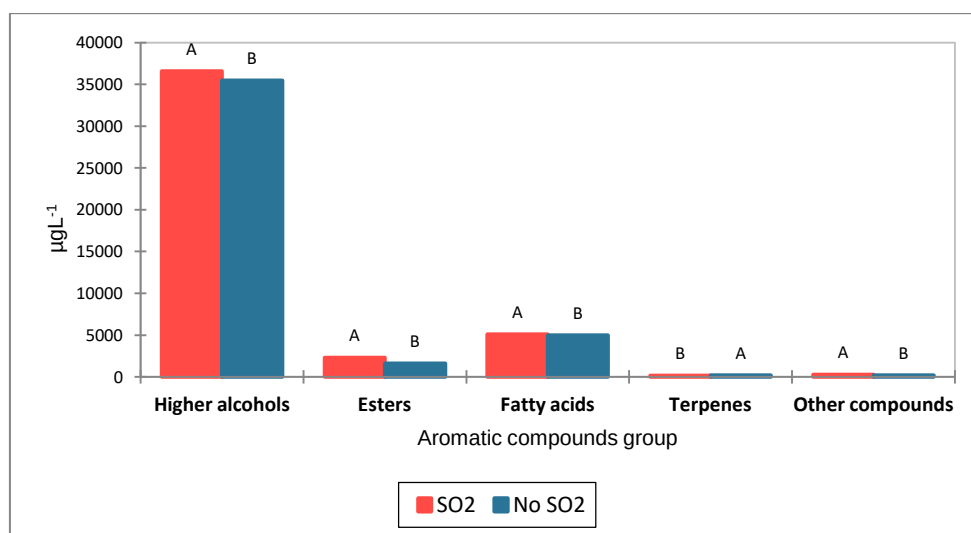
high total acidity. The ash content in the wine was low and did not meet the criteria for quality wines ($> 1.4 \text{ g L}^{-1}$).

Table 1. The basic physicochemical composition of Graševina wines

Parameter	With SO ₂	No-SO ₂
Specific gravity	0.9888	0.9884
Alcohol (vol %)	15.5	15.2
Total extract (g L ⁻¹)	21.9	20.1
Reducing sugar (g L ⁻¹)	3.4	1.6
Total acidity(g L ⁻¹)	9.1	8.5
Volatile acidity (g L ⁻¹)	0.38	0.35
pH	3.07	3.11
Ash (g L ⁻¹)	1.22	1.31
Free SO ₂ (mg L ⁻¹)	23.0	0
Bound SO ₂ (mg L ⁻¹)	93.0	0
Total SO ₂ (mg L ⁻¹)	116.0	0

As it can be seen in Table 1. treatment with SO₂ contained an optimal concentration of sulfur dioxide, which is within the permissible limits, while the no-SO₂ treatment had no sulfur dioxide at all, which was the aim of this research.

Aromatic composition of wine



Different upper case letters in the same column represent statistically significant differences between means at $p \leq 0.05$

Fig.1. Concentration of aromatic compound groups ($\mu\text{g L}^{-1}$) in Graševina wines

In this study, 75 aromatic compounds were detected in wines produced with SO₂ and 56 compounds were detected in wines without SO₂. All compounds were divided into six aromatic groups: higher alcohols, esters, fatty acids, terpenes, C13-norisoprenoids and other compounds. According to ANOVA analysis, significant differences ($p < 0.05$) were noticed for all groups of aromatic compounds, except for C13-norisoprenoids.

Higher alcohols were the most abundant group of aromatics with a concentration of about 35 mg L⁻¹. They have a positive effect on wine flavor when concentrations are below 300 mg L⁻¹. Six higher alcohols were detected in this analysis, four of which were above the detection threshold including: 1-decanol, 1-dodecanol, 1-propanol and (6Z)-nonen-1-ol, which was found only in wine made with SO₂. It is important to note that the alcohols above the detection threshold were more prevalent in wines with SO₂. These alcohols contribute to the pear, coconut, wax, ripe fruit, melon and green aromas.

Four fatty acids were detected in this study: isovaleric, decanoic, hexanoic and octanoic acid. Their concentration was higher in wine with SO₂ and all of them were above the detection threshold. They enrich the wine with cheesy, goaty, oily and waxy aromas. In high concentrations they can be undesirable.

Esters were present in higher concentrations in wines with SO₂ and five of them were found in wine: 2-phenylethyl acetate, isoamyl acetate, ethyl decanoate, hexanoic acid and ethyl-9-decanoate, all above the detection threshold. They contribute to the wine with aromas of rose, banana, honey, green apple and chamomile.

Sensory analysis of wine

The sensory evaluation was carried out by six certified evaluators from the Faculty of Agriculture in Zagreb. Each evaluator used scores from 0 to 5 to express the intensity of a particular sensory property (0 to 1 - weakly pronounced, 2 to 3 - moderately pronounced, 4 to 5 - strongly pronounced).

In figure 2. the evaluated parameters and their expression can be seen. Based on the average score of the overall impression, it can be seen that Graševina with SO₂ was rated as better. However, the average score of both samples was relatively low, indicating the moderate quality of wines.

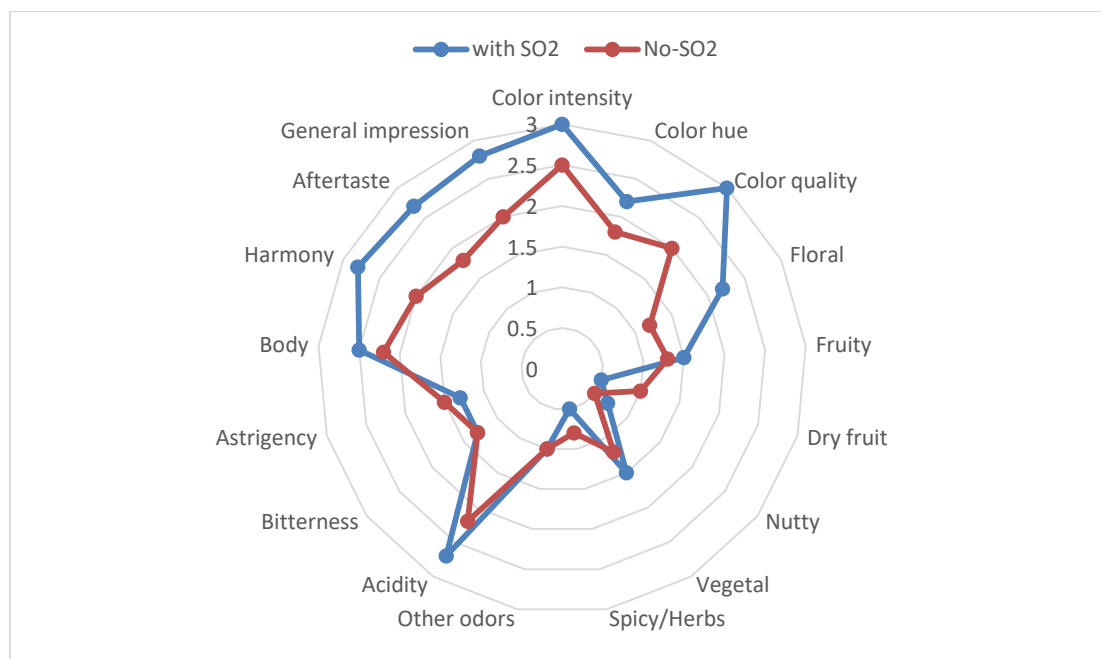


Fig 2. Descriptive sensory evaluation of Graševina wines with and without SO₂

CONCLUSION

In this study, the influence of alternative reducing and antimicrobial agents in comparison to SO₂ treatment on Graševina white wine quality was analyzed. More aromatic compounds (75) were detected in Graševina wines produced with SO₂ in comparison to wines with glutathione and alternative reducing agents such as glutathione and inactive yeasts (56). Six higher alcohols and four fatty acids together with total esters were found in higher concentration in wines with SO₂. Significant differences were found among all analyzed groups of aromatic compounds regarding the treatment, except for C13 norisoprenoids. According to the physicochemical analysis it can be stated that no significant differences were found in the composition of Graševina wines. According to the concentration of sugar-free extract, both wines met the criteria for the premium category, but not according to the ash concentration.

Both wine samples were sensory rated as mediocre regardless the treatment applied, which was also due to the unbalanced composition of grapes. According the sensory evaluation Graševina wine produced with SO₂ was declared as better quality wine. In order to draw more reliable conclusions about the impact of alternative protocols for wine protecting on the quality and composition of white wines, it is necessary to repeat the research during several vintages.

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ANALYSIS OF HEMATOLOGICAL PARAMETERS OF GOATS IN THE AREA OF THE AUTONOMOUS PROVINCE OF VOJVODINA (REPUBLIC OF SERBIA)

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SUMMARY

Determination of hematology parameters is important for monitoring the health status. The research was conducted on 20 goats. The aim of the work was to determine hematological parameters and to compare them with the values of other authors. Since there are no papers on this topic in the Republic of Serbia, these results will be useful to vets dealing with goat health. There is space in future to determine these hematological parameters on a larger number of samples, in order to obtain an even more precise picture and to be able to create reference values for the examined geographical area.

Key words: hematology, erythrocytes, leukocytes, Vojvodina, goats

INTRODUCTION

Animal husbandry is changing more and more rapidly today, in order for the producer to be successful, it is necessary to constantly follow the progress in the field of science that would help him in the best and highest quality production. Goat farming as a field is no exception to this rule, with many producers starting their careers in it (Bogdanović et al., 2011). Serbia has excellent potential when it comes to goat breeding, good geographical location, soil composition, appropriate flora and fauna, therefore goat farming in Serbia is increasingly prevalent (Mekić et al., 2014). Goats provide high-quality meat and milk products, and are considered ideal animals for breeding due to their high degree of adaptation and resistance (Al-Bulushi et al., 2017). The quality of goat products, especially milk, is evidenced by a comparison of commercial goat milk with cow's milk, research has shown that goat's milk has more nutritional value than cow's milk. Namely, a better ratio of Na and K was observed, as well as a higher concentration of fatty acids, which have a positive impact on human health. Also a large amount of I (iodine) is a positive side, due to the increasing deficit in the world (in adults). A lower concentration of saturated fats in goat's milk has also been shown, which is also suitable for humans due to the reduction of the risk of cardiovascular diseases. In addition, cow's milk also contains higher concentrations of Ca, S, Zn, omega-3 and lignans (Stergiadis et al., 2019). The Alpine goat originates from the Swiss-French Alps, spread all over the world, it is considered one of the best noble dairy breeds.

Determination of hemato-biochemical parameters of small ruminants is important for monitoring the metabolic and health status of animals. Race, gender, age category, diet, season of the year, lactation period, altitude have a great influence on the hematological picture (Antunović et al., 2019).

The Autonomous Province of Vojvodina is located in the Pannonian plain within the Republic of Serbia. It is located in the northern part of the Republic of Serbia at a low altitude. The climate in Vojvodina is moderately continental with a distinct transition through all seasons, summers are warm and winters are cold, while autumn and spring are short (Katic et al., 2008). As certain hematological parameters depend on the area where goats are raised, there is insufficient data on the hematological parameters of goats in our conditions of keeping, in the territory of the Republic of Serbia, and especially in the plains of Vojvodina. Therefore, research should be carried out in this direction in order to obtain new data that will be of great importance to clinical veterinarians dealing with goats. In this way, the health condition of the goats will be improved and potentially better economic results will be obtained (Antunović et al., 2013).

The aim of this study is to determine certain hematological parameters in healthy lactating goats, which are bred on the territory of the Autonomous Province of Vojvodina (Republic of Serbia) and compare them with the results of other researchers, in order to obtain values that are physiological in our keeping conditions.

MATERIAL AND METHODS

In this experiment, goats of the Alpine breed (French Alpine) were used. The research included 20 goats. All goats were female, in the lactation phase, and a venipuncture was performed on them at vein jugularis in order to obtain material for testing. Blood sampling was done in the early hours of the morning. The experiment was carried out on 4 agricultural farms in the territory of the Autonomous Province of Vojvodina. The collected material was examined in the laboratories of the Department of Veterinary Medicine, which is part of the Faculty of Agriculture, University of Novi Sad.

Hematological parameters - number of erythrocytes (RBC); hemoglobin (HGB); hematocrit (HCT); mean corpuscular hemoglobin (MCH); mean corpuscular hemoglobin concentration (MCHC); mean corpuscular volume (MCV) and total leukocyte count (WBC) were determined on an automatic hematology analyzer manufactured by Nihon Kohden model MEK-6550 (Tokyo, Japan) for veterinary use. Differential blood count (leukocyte count) was performed by manual method using a microscope manufactured by BioBlue model BB.4253 (Arnhem, The Netherlands). A total of 20 blood smears were examined, which were stained using the BIO-DIFF RTU KIT staining kit (Biognost, Zagreb, Croatia) for leukocyte differentiation. During staining, we followed the manufacturer's instructions given in Table 1. A total of 100 leukocytes were identified on each blood smear, which is why each type of leukocytes was presented in percentage values. After that, the absolute value of individual leukocytes was calculated using the following formula.

$$\text{Number of specified leucocytes} = \frac{\text{Total number of leucocytes} \times \text{Percentual participation}}{100}$$

Statistical analysis conducted using MS Office package.

Table 1: Protocol regarding coloring blood stains with the *BIO-DIFF RTU KIT*

Nr.	Coloring phase	Duration
1.	Let the bloodstain dry	
2.	Dip the smear in Bio-Diff 1 RTU reagent	5 x 1 second
3.	Drain the excess reagent from the plate onto filter paper	
4.	Dip the smear in Bio-Diff 2 RTU reagent	3 x 1 second
5.	Drain the excess reagent from the plate onto filter paper	
6.	Dip the smear in Bio-Diff 2 RTU reagent	6 x 1 second
7.	Rinse the bloodstain with a pH 6,8 buffer	1 minute
8.	Letting the blood stain dry	

RESULTS AND DISCUSSION

The most common factors affecting hematological and biochemical parameters can be age, race, weather, sex, altitude, disease, stress, diet (Karaşahini et al., 2022). In our study, the analysis was performed on 20 clinically healthy lactating goats in plain areas. All the hematological results obtained in this study are presented in Table 2, as well as graphically for the leukocyte formula (Figure), for easier visual inspection of the percentage representation of leukocytes. Erythrocytes are blood cells, known as red blood cells (RBC). The main function of erythrocytes is the transfer and supply of tissues with oxygen, as well as the transport of carbon dioxide from the tissues to the lungs (Kendal et al., 2010). When processing our results from a total of 20 samples, the number of RBC ranged from 7,8 to 16,5x10¹²/l. Researchers Antunović et al. (2013) come to the result that the values for RBC ranged from 12,5 to 14,74x10¹²/l during lactation. In a study by researchers Arfuso et al. (2016) made a comparison by age categories and by different breeds of goats, where the results ranged from 11 to 25x10¹²/l, which is in agreement with our results. Another study that obtains results that are in line with our results is the study by researchers Antunović et al. (2019). Researchers Bezzer et al., (2008) found a mean value of 14,48x10¹² while the min/max range was 8,6 to 20x10¹². The paper did not define race or age category, but there was a larger number of samples than in our study (n=143). The results that are not in agreement with our results are the results of researchers Dhuha et al. (2021). Namely, in this

study the results obtained for the number of RBC ranged from 3,78 to 4,61x10¹²/l. One of the possible reasons for this disagreement is that these researchers performed blood sampling throughout the whole year, while in our study, sampling was performed in the spring.

Table 2: Hematological results

Studied parameter	Sample number (n)	Average	SD (±)	MIN	MAX
Erythrocyte number (x10 ¹² /l)	20	12,98	2,58	7,8	16,5
Hemoglobin concentration (g/l)	20	102,49	8,81	83	123
Hematocrit (%)	20	26,25	4,51	19,7	34,1
Total leucocytes number (x10 ⁹ /l)	20	11,48	2,80	6,7	18,2
Neutrophiles (%)	20	42,3	9,95	23	59
Eosinophiles (%)	20	1,2	1,94	0	6
Basophiles (%)	20	2	2,49	0	9
Lymphocytes (%)	20	53,2	9,21	40	73
Monocytes (%)	20	1,30	2,58	0	10
MCV (fL)	20	20,60	3,06	15,27	26,41
MCHC (g/dL)	20	39,84	5,75	31,09	50,67
MCH (pg)	20	8,15	1,43	6,54	12,18

Leukocytes are known as white blood cells (WBC), they are blood cells whose main role is to defend the body against various pathogens. They can be divided into granulocytes and agranulocytes. Granulocytes are neutrophils, eosinophils, and basophils, while agranulocytes are lymphocytes and monocytes (Bhardwaj et al., 2021). The total number of WBC in our study ranged from 6,7 to 18,2x10⁹/l. Lymphocytes were the most abundant WBC in the blood (53,2%), followed by neutrophils (42,3%), basophils (2%), monocytes (1,3%) and eosinophils (1,2%), which were the least abundant in the blood (Graph). These results for the WBC are in agreement with the results of researchers Antunović et al. (2013) and Arfuso et al. (2016), while the results for the differential blood count were not given in the mentioned studies, we cannot make that comparison. In a study by researchers Dhuha et al. (2021)

results for the leukocyte formula are in agreement with ours, except in the case of the percentage of lymphocytes, which are from 35,53 to 42,80%, while in our study the average value for the percentage of lymphocytes was 53,2%. We can consider that our values are more "appropriate" because goats physiologically have a lymphatic blood picture, i.e. the most abundant leukocyte is a lymphocyte, which matches our results. Also minor deviations related to the leukocyte formula were recorded between our results and the results in the studies of researchers Antunović et al. (2019) and studies by Bezzera et al. (2008). The first study had a slightly lower percentage of neutrophils, while the second study had a slightly higher percentage of neutrophils compared to our results.

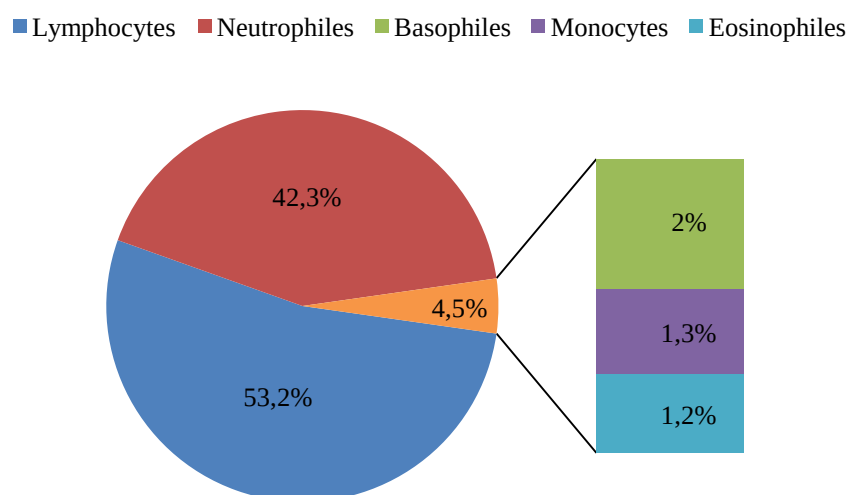


Figure: Leucocyte percentages (leucocyte formula)

Hemoglobin is a metalloprotein, which is found in erythrocytes, and has a key role in the processes of binding and transporting oxygen (Kendal et al., 2010). Hematocrit represents the percentage of blood that is made up of erythrocytes, the percentage ratio of the liquid and cellular part of the blood (Thrall et al., 2012). For hemoglobin concentration values in our study, the mean value was 102.49 g/l, while the hematocrit value was 26,25%. The values for hemoglobin concentration in all mentioned authors (Antunović et al., 2013; Arfuso et al., 2016; Antunović et al., 2019; Dhuha et al. 2021; Bazzera et al. 2008) are in agreement with our results. The hematocrit value was higher in the results of the study by Dhuha et al., (2021) compared to ours and was 41,40% and decreased in the study by the group of authors Karaşahin et al., (2020) and was from 21,29 to 22,67%.

Hematological indices represent blood parameters that are determined when determining the blood count and are made up of MCV (mean corpuscular volume), MCHC (mean corpuscular hemoglobin concentration) and MCH (mean corpuscular hemoglobin) (Naeim et al., 2013).

Our MCV results were 20,60 fL. In the study by researchers Antunović et al., (2013), values for MCV of 28,01 to 29,48 fL were recorded, which represents higher values compared to our study. In a recent study by researchers Antunović et al., (2019) the

comparison is similar, values for MCV from 33,69 to 62,71 fL are observed. Researchers Arfuso et al., (2016) in their study reach results for MCV of 8,9 to 30,0 fL, which is in agreement with our results. Our results for MCV are not in agreement with the results obtained by researchers Dhuha et al., (2021), where an increase in the MCV index was observed, and in their study it ranged from 110,28 to 112,80 fL.

The mean value for the MCHC index was 39,84 g/dl. This value was slightly elevated compared to the reference values reported in the "Merck" Veterinary Manual (MSD, 2022). Reference values in this manual are in the range of 30 to 36 g/dl. Similar values as in the "Merck" Veterinary Manual (MSD, 2022) are also stated in their study by researchers Antunović et al. (2013). Values for MCHC in the study by Arfuso et al., (2016) ranged from 25,7 to 85,0 g/dl, which is in agreement with the results of our study, while researchers Antunović et al., (2019) and Dhuha et al., (2021) obtained values that are lower than our obtained results.

The values of the hematological index MCH in our study were 8,15 pg, which represents slightly elevated values compared to the "Merck" Veterinary Manual (MSD, 2022), which states reference values of 5,2 to 8,0 pg. Our results are in agreement with the results of studies by Antunović et al., (2013) and Arfuso et al., (2016). In a more recent study by Antunović et al. (2019), agreement was also noted in the results for MCH.

Compared to the "Merck" Veterinary Manual (MSD, 2022), our results are within the physiological values for the largest number of examined parameters (number of erythrocytes, hemoglobin concentration, hematocrit, total number of leukocytes, percentage ratio of neutrophils, eosinophils, basophils, lymphocytes, monocytes, and for MCV index).

CONCLUSION

The results obtained in our study are within the reference values both in comparison with the "Merck" Veterinary Manual (MSD, 2022) and with the results of most other authors. Of course, minor deviations were observed in some parameters from the results of other authors, however, the number of hematological parameters is affected by factors such as age, sex, race, time of year, temperature, altitude, diet, or even the possibility of pathological processes at the time of blood sampling. As there are no papers on this topic in the Republic of Serbia, these results will contribute to a better understanding and will be useful primarily to clinicians dealing with goat health as possible physiological frameworks for hematological parameters. There is space in future research to determine these hematological parameters on a larger number of samples, in order to obtain an even more precise picture and to be able to create reference values for the examined geographical area.

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SYNOVIAL FLUID CHARACTERISTICS OF OSTEOARTHRITIC STIFLE JOINT IN DOGS

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SUMMARY

Osteoarthritis is undoubtedly one of the most common joint diseases in dogs. We distinguish between primary and secondary. The latter is also much more frequent, resulting from numerous causes such as hip dysplasia or anterior cruciate ligament rupture. This degenerative disease is divided into 4 basic grades depending on the extent of changes affecting the joint, from the first to the fourth. The aim of this research was to determine 4 parameters of synovial fluid obtained by arthrocentesis of a stifle joint with a previously determined degree of osteoarthritis. 24 dogs of medium-sized breeds who were also patients of the Clinic for Surgery, Orthopedics and Ophthalmology, Faculty of Veterinary Medicine University of Zagreb, participated in the research. Approval for this research was obtained from the Faculty Ethics Committee. They were divided into 4 groups depending on the X-ray-determined degree of osteoarthritis, and each group consisted of 6 dogs or synovial fluid samples. The physical characteristics were determined, namely volume and color, as well as the chemical accuracy of glucose and lactate. Physical analyses were performed immediately after knee puncture, while chemical analyses were performed at the Department of Pathophysiology, Faculty of Veterinary Medicine, University of Zagreb. The results showed that an increase in the degree of osteoarthritis leads to an increase in the sample volume. The color of the samples varied from colorless, light yellow to red. A slightly darker yellow color was observed in certain samples belonging to the groups with 3rd and 4th degree osteoarthritis. At the same time, lighter shades of synovium were present mainly in samples with lower degrees. An increase in lactate concentration with an increase in the degree of osteoarthritis has also been proven. Glucose concentration, however, does not differ significantly between individual degrees of osteoarthritis. The results of the tested parameters in this study also confirm that with an increase in the degree of osteoarthritis of the stifle joint, there are also significant changes in the composition of the synovial fluid itself.

Keywords: Osteoarthritis, Stifle joint, Synovial fluid, Arthrocentesis

INTRODUCTION

Osteoarthritis is a chronic, degenerative and progressive disease of the joints, accompanied by thinning of the joint capsule, degeneration of the joint cartilage, thickening of the subchondral bone, and the inflammatory process of the synovial membrane. The changes as mentioned above eventually lead to the appearance of clinical signs such as limping, joint stiffness, and reluctant activity. During palpation and manipulation of the affected joint, we often encounter swelling, pain, and crepitation. Indeed, the extent of clinical signs depends on the degree of osteoarthritis of the affected joint. We distinguish 4 basic grades of osteoarthritis according to the Kellgren Lawrence system for classification of osteoarthritis (Figure 1). In grade I, there is a preclinical phase. Numerous predisposing factors will favor the development of this initial stage of osteoarthritis. Such factors can be divided into two basic groups, local and systemic. The main systemic factors are age, obesity, breed and genetic predisposition. Local factors primarily include any form of trauma such as rupture of the anterior cruciate ligament, developmental anomalies such as hip dysplasia, surgical procedures, muscle weakness, etc. (5). In the mentioned phase, due to the increased metabolism of chondrocytes, cartilage matrix begins to be degraded by the action of metalloproteinases. Osteoarthritis of the second degree is considered mild. The level of collagen and proteoglycan within the synovial fluid increases. Also, during this phase, bone erosion begins due to cartilage lesions, which can lead to the growth of a new bone and the formation of osteophytes. The third grade of osteoarthritis is described as moderate. It is characterized by increasingly pronounced symptoms and the development of stronger inflammation. Thinning of the cartilage occurs, thereby reducing the interosseous space. In this phase, the presence of synovial macrophages is increasingly pronounced. The last stage, and at the same time the most severe, is stage 4 of osteoarthritis, which is accompanied by almost complete loss of cartilage, and the joint's mobility is significantly reduced. In addition to primary or idiopathic osteoarthritis, there is a secondary one, which is much more common. The most common secondary cause of knee osteoarthritis in dogs is a rupture of the anterior cruciate ligament (4). Predisposed breeds for this condition are Labradors and Rottweilers (3). It is believed that the genetic predisposition of the aforementioned breeds, as well as the thinner fibrous ligament fibers in Labradors, favor ligament rupture.

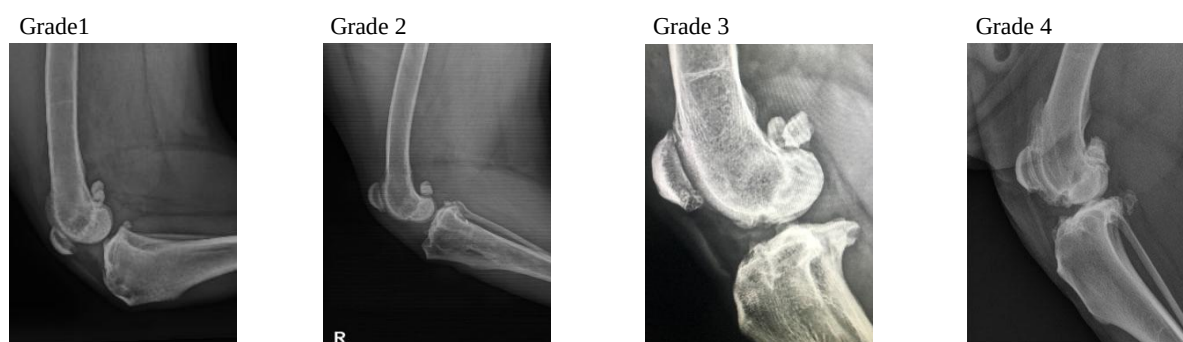


Figure 1. Kellgren-Lawrence radiographic grading criteria for osteoarthritis (Authors picture, 2022.)

As a result of the onset of osteoarthritis, there are inevitable changes in the amount and composition of the synovial fluid of the associated joint, the extent of which varies depending on the degree of inflammation. The otherwise clear, colorless to light yellow, viscous, avascular, and hypocellular fluid undergoes changes during the inflammatory process. The synovium has multiple roles; first of all, it acts as a natural lubricant and a biochemical pool through which numerous nutrients and regulatory cytokines are transported to the surrounding tissues (6). The aim of this research was to compare the composition of 24 different samples of synovial fluids obtained by arthrocentesis of the knees of dogs with different degrees of osteoarthritis.

MATERIALS AND METHODS

As part of this research, in the period from 01.02.2022. until 01.07.2022. a population of 24 dogs, also patients of the Clinic for Surgery, Orthopedics and Ophthalmology, Faculty of Veterinary Medicine University of Zagreb, were examined. Approval for this research was obtained from the Faculty Ethics Committee. All dogs were medium breeds, weighing 15 to 20 kg, aged 3 to 12 years old, and both sexes. Historically, dogs were associated with hind limb lameness of varying duration and intensity. With regard to the X-ray-determined degree of osteoarthritis, dogs were divided into groups by degree of osteoarthritis so that each group consisted of 6 samples of synovial fluid. Altogether, a total of 4 groups were made according to 4 grades of osteoarthritis. Stifle joint synovium samples were taken by puncturing the joint with a 20 G needle (Figure 2). Patients were placed in the lateral position, and the knee joint was approached from the craniomedial side, directing the needle distally from the patella and laterally from the patellar ligament. Synovial fluid was then aspirated using 2 ml syringes.



Figure 2. The punctuation of a stifle joint and aspiration of synovial fluid using a 20 G needle (Authors picture, 2022.)

The physical properties of the sample, precisely the volume and color, were determined immediately after the puncture. The resulting liquid was placed into Eppendorf tubes without anticoagulants of a volume of 1 mL for biochemical analysis. Synovial fluid samples were transported to the laboratory of the Department of Pathophysiology at room temperature immediately after sampling, i.e. within 30 minutes, in order to prevent cell decomposition and changes in biochemical parameters. Further chemical analyses, i.e. determination of glucose and lactate concentrations of the samples taken, were carried out at the Department of Pathophysiology of the Faculty of Veterinary Medicine. The chemical analysis of the synovial fluid included the determination of lactate, which was determined within 2 minutes after sample collection using a manual and commercially available lactate meter (Lactate-Scout: Senslab, Leipzig, Germany), and the determination of glucose by the GOD-PAP method (Glucose liquicolor, Human GmbH, Germany) with a spectrophotometer (Helios Gamma, Thermo electron corporation). The concentration of glucose in the samples was obtained using the formula. The ANOVA test was used to compare lactate and glucose concentrations between the four groups (Table 1.)

Table 1. Results of statistical analysis of lactate and glucose concentration in synovial fluid using ANOVA test.

Variable	F	p-value
Lactate	8,917	0,009
Glucose	0,321	0,811

RESULTS AND DISCUSSION

In a period of 5 months, the physical and chemical characteristics of synovial fluid samples taken by arthrocentesis of the knee joint in 24 dogs were examined. A certain degree of osteoarthritic changes in the knee joint was determined radiographically in all samples. The samples were divided into 4 groups according to the degree of osteoarthritis. Each group consisted of 6 samples of dogs' synovium of the knee joint. The average amount of all obtained volumes of synovial fluid samples was 0.9 ± 0.38 ml. A larger amount of synovial fluid was found when sampling knees with a higher degree of osteoarthritic changes (grades 3 and 4), while in knees with a mild degree of osteoarthritis, smaller volumes of synovium were obtained (groups 1 and 2). The average volume of samples of the first group (first degree of osteoarthritis) was 0.6 ± 0.1 ml. The volume of 0.8 ± 0.2 ml was determined in the group containing only samples with established 2nd degree osteoarthritis. The last two groups with a higher degree also showed significantly larger volumes of synovium obtained with average values of 1.1 ± 0.3 ml in the 3rd group and 1.3 ± 0.15 in the last group with the 4th degree of osteoarthritis. According to previous research, an increase in synovial fluid volume is an expected phenomenon in inflammatory conditions affecting the joint (1). The color of aspirated synovium varied from physiological light yellow to red. The red discoloration was

found in 10 samples, which can be attributed to contamination with blood during the puncture of the knee. In other samples, light yellow synovium was observed, of which as many as 5 belonged to the first two groups. The other 9 samples belonged to groups with a higher degree of osteoarthritis with slightly darker and less transparent synovium. The results showed that lactate concentrations in the 3rd and 4th group were significantly different from the other two groups (p-value=0,009). Higher concentrations were observed in the group with the third (6.8+-2.15 mmol/dl) and fourth (9.2+-0.17 mmol/dl) grade of osteoarthritis, compared to the first (2.9+-0, 8 mmol/dl) and second (3.2+-0.14mmol/dl) grade (Figure 3). Previous research indicates an increased concentration of lactate in septic arthritis in dogs, which can be associated with a higher degree of osteoarthritis (7).

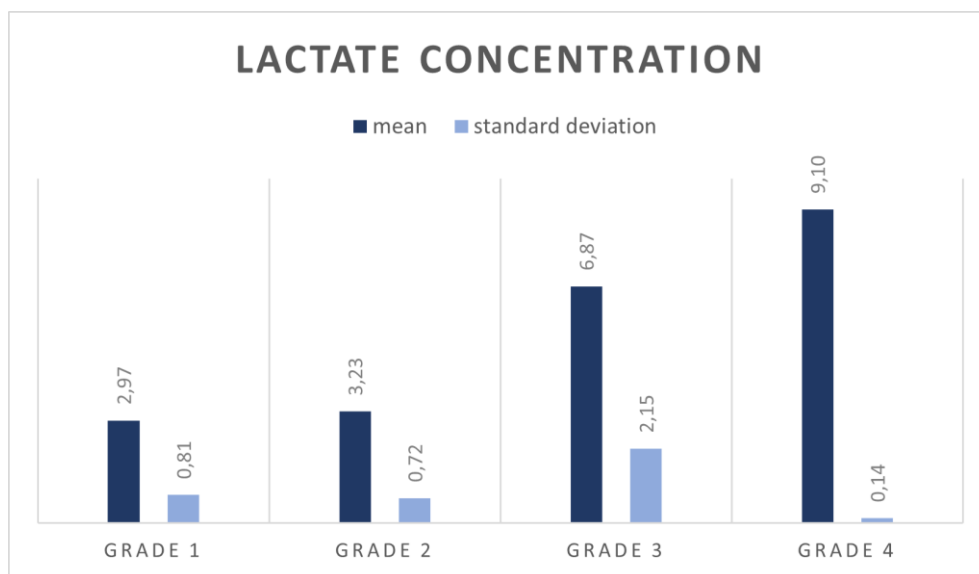


Figure 3. The concentration of lactate in synovial fluid observed in 4 grades of osteoarthritis (Authors picture, 2022.)

The increased concentration of glucose (5.16+-2.27mmol/dl) in our samples corresponds to the results of previous studies where the synovial fluids of healthy and osteoarthritis affected knees were compared (2). Still, it does not differ significantly between individual degrees of osteoarthritis (p-value=0,811) (Figure 4).

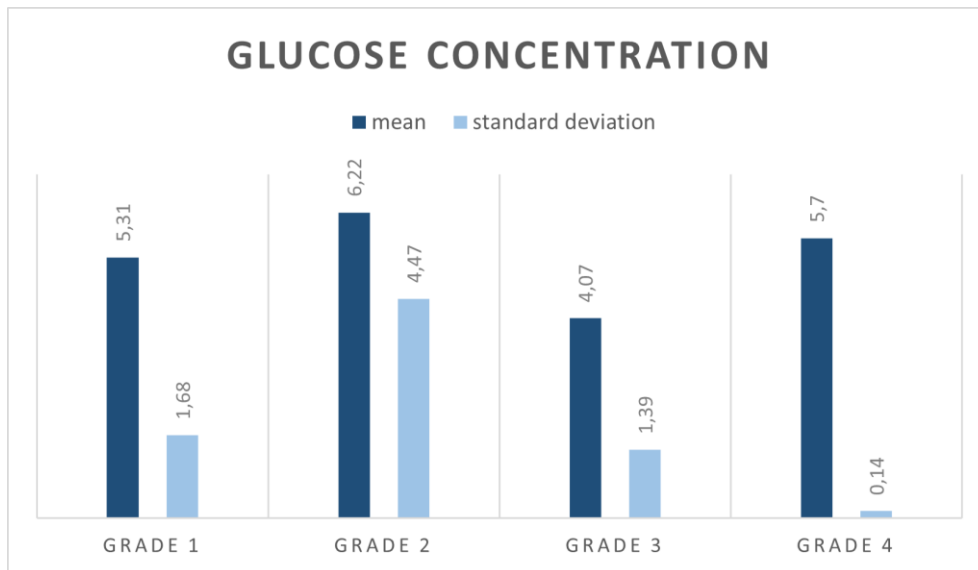


Figure 4. The concentration of glucose in synovial fluid observed in all 4 grades of osteoarthritis (Authors picture, 2022.)

CONCLUSION

This study showed differences in the degrees of osteoarthritis in the physical and chemical properties of the synovial fluid of a dog's knee with a rupture of the anterior cruciate ligament. Thus, the increased volume of synovial fluid is proportional to the degree of osteoarthritis and darker color of the synovial fluid, while transparency decreases in samples with a higher degree of osteoarthritis. The lactate concentration also increases as the degree of osteoarthritis increases, which can be attributed to potential septic conditions within the joint. Lactate can therefore be a potential indicator of septic arthritis in diagnostics, but it is necessary to perform a cytological analysis of the sample to confirm the diagnosis. Due to the increased glucose concentration in all samples, we do not consider glucose in this study to be a suitable parameter for determining the degree of osteoarthritis.

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BUFFER ZONES AS A SOLUTION FOR POLLUTION IN AGRICULTURAL LANDSCAPE. CASE STUDY: SUGAR FACTORY IN BAČ, SERBIA

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SUMMARY

„Sunoko“ is a Sugar Refining Mill located in Bač that processes sugar beet and produces sugar. Around the factory there is a buffer zone that protects the nearby agricultural land and the DTD canal from pollutants. Field observation provided data on the current state and effectiveness of the buffer zone and the adaptability of existing tree and shrub species to environmental, climatic and soil conditions. The goal of this paper is to emphasize the importance of buffer zones around factories in agricultural landscape in order to establish a more balanced ecosystem and improve the quality of life.

Key words *agricultural landscape, buffer zone, tree adaptability, stabile ecosystem, sugar factory*

INTRODUCTION

According to Ebregt and De Greve (2000), a buffer zone is an area which lies between two or more other areas and serves to reduce the possibility of damaging interactions between them. Buffer zones are areas of land under permanent vegetation, which helps control the quality of water, air and soil, but also helps solving problems that occur in agricultural practice. Buffer zones contribute to enhanced filtration of nutrients and pesticides and prevent their runoff into local surface or underground waters (Baumhardt & Blanco, 2014). Although buffer zones are capable of removing pollutants from surface runoff, the proximity of buffer zones to sources of pollution is more important to their effectiveness. A buffer zone must be comprehensively arranged along streams and around pollution sources to be successful. The value of buffer zones is reflected not only in their ability to control water quality but on several other benefits and costs associated with maintaining large areas of natural vegetation (Noris, 1993).

The use of buffer zones for removing sediment and other pollutants from runoff is a relatively new practice. Decades of industrial activity have caused serious soil pollution in thousands of locations in cities around the world, posing a risk to human health as contaminants can reach food or water through agricultural practices (Song et al., 2019). In agricultural landscapes, buffer zones often consist of a fenced area alongside streams that stock is excluded from and this may be left as a grassy sward, or planted with woody

vegetation (Parkyn, 2004). Buffer zones have great potential for the control of water-based pollution and the conservation of biodiversity; they should, as a consequence, be applied more widely (Haycock et al., 1996).

Buffer zones can be erected for phytoremediation (the use of plants and their associated microorganisms in the purification of the environment) in the area of factory plants, industrial zones, watercourses, etc. Phytoremediation has ecological and socio-economic advantages (Yadav et al., 2018). Phytoremediation is a practice that uses plants to degrade the contaminants from soil and water (EPA, 2000). It has a low cost, both in terms of the initiation process itself, as well as in terms of area maintenance. It represents the cheapest technology for removal of heavy metals from the soil, because it is 60-80% cheaper compared to conventional physico-chemical techniques (Mwegoha, 2008). As a "green technology" with the potential to remove, store or transport pollutants (heavy metals, organic pollutants) from the soil with minimal generation of secondary waste, it represents a sustainable way of soil remediation (Delvert, 2017). Buffer zones intended for phytoremediation enable easier and safer agricultural production. Industrial plants are often surrounded by arable land, so the harvested crops must be healthy and safe to use. The advantages of buffer zones for farmers are also reflected in the reduction of damage from floods, erosion, the protection of the runoff of pollutants from the surrounding industry into the arable land, and thus the improvement of the soil quality.

The goal of this study is to analyze the vegetation of the buffer zone within the complex of the "Sunoko" Sugar Refining Mill in Bač. The analysis is carried out to determine the state of the existing vegetation. Based on the obtained results, the functionality of the buffer zone, and adaptability of existing species to environmental, climatic and soil conditions will be determined. This work would emphasize the importance of raising buffer zones in agricultural and industrial landscapes and show the necessity of their establishment in order to improve air, land and water quality which will create a more sustainable and healthier environment.

MATERIAL AND METHODS

Study area

The complex of "Sunoko" Sugar Refining Mill is located in Bač, on the west of Vojvodina province in the northern part of Serbia. The Bač municipality is located in Vojvodina, in the South Bačka district. It covers only 365 km², of which 28,198 ha is agricultural and 4,775 ha is forest land. The territory of the Bač municipality borders the municipalities of Bačka Palanka and Odžaci on the north, northeast, east and southeast sides, while on the west side the border is the Danube river, on whose west bank is the Republic of Croatia. The complex of "Sunoko" Sugar Refining Mill is located on the outskirts of town Bač, on the Bač-Bačka Palanka road. Next to the factory complex, there are two important landscape elements the Danube-Tisa-Danube canal (DTD canal) and the area of the Special Nature Reserve "Karađorđevo" (Figure 1). The surrounding land is used as arable land. The factory complex covers an area of approximately 50 ha. This factory was founded in 1975. Until the end of 90s, it operated as a state-owned factory, before being privatized in the early

2000s. While operating at full capacity, the factory processed 4,000 tons of sugar beet per day. During 2003 and 2004, the annual production was 17,700 tons of sugar. The factory still operates today, but with a significantly reduced capacity (Internet source 1).



Figure 1. Study area
(Source: GeoSrbija, modified by authors, 2022)

Data collection and processing

The field observation was followed by an analysis of the literature and the collection of data on hydrology, pedological and climate characteristics in the territory of the Bač municipality (Strategy of socio-economic development of the municipality of Bač 2009-2014, 2009). This was followed by an evaluation of the state of the individuals to determine their adaptability to the existing conditions of the location. The buffer zone in the area of sugar factory in Bač covers a little more than 6 ha. Determination of species and evaluation of the conditions of individuals were done by field observation and using the appropriate literature (Cvjetičanin et al., 2016). The share of each plant species within the protective zone was determined. Taking into account that the buffer zone is planned, and that the species within the zone are of the same age class, the characteristics of each plant species are shown as the average value of several individuals. The following characteristics of plant species within the buffer zone are shown: height, height of trunk to the first branches, crown width, trunk diameter, rotting of the trunk and thicker branches, dried branches, logging of thick branches, dried crown top, mechanical and phytopathological damage (Anastasijević, 2007). Due to the lack of instruments, all the processed data were collected by the visual method on field. The obtained data were processed in the Microsoft Office Excel software. The necessary measurements were made through the application of the Google Earth tool for measuring distances and areas.

RESULTS AND DISCUSSION

Based on the collected data on hydrological characteristics, the results show that the municipality of Bač has very developed hydrology. Surface streams cover the territory of the entire municipality (Strategy of socio-economic development of the municipality of Bač 2009-2014, 2009). The Danube-Tisa-Danube canal, the most important hydro system in Vojvodina, is located near the sugar factory, i.e. the researched buffer zone, so the factory has a direct influence on it. On the territory of the Bač municipality, the depth of groundwater is not uniform but depends on the proximity of the Danube, the geological base, as well as the geomorphological features of the terrain. The depth is the smallest in the alluvial plane, where it practically emerges on the surface of the terrain. On the loess terrace the level of groundwater oscillates at a greater depth and tends to move towards the alluvial plane (Strategy of socio-economic development of the municipality of Bač 2009-2014, 2009). As the settlement of Bač (and thus the buffer zone) is located at the contact of the loess terrace and the alluvial terrace, the groundwater level is not high, which is suitable for plant species within the buffer zone. In terms of pedological composition, the municipality of Bač is atypical in the region of southern Bačka, because it has a smaller proportion of chernozem compared to municipalities surrounding the area (Strategy of socio-economic development of the municipality of Bač 2009-2014, 2009). As already mentioned, the buffer zone in the area of the sugar factory is located at the contact of loess and alluvial terrace, and there are several types of soil in that area. The species of the buffer zone do not show great demands on the soil, and they thrive in this area. Almost all the species present in the buffer zone thrive on very different soils, so they have developed well in this place as well. Based on the collected data on the climatic characteristics of the municipality of Bač, it can be said that the climatic conditions correspond to the development of most plant species. Given that the weather conditions are quite stable throughout the year, i.e. that the summers are marked by an average temperature of 28.3°C and the alternation of warm and cooler weather, and that the temperature of the winter months does not drop below -3°C on average (Strategy of socio-economic development of the municipality of Bač 2009-2014, 2009), very good conditions for the growth and development of many plant species are created. However, the existence of tropical days in the summer and frosty days in the winter are not foreign to this area. Therefore, all plant species must be selected in such a way that they are as resistant as possible to such extremes. Also, the higher amount of precipitation in this part of Vojvodina contributes to better adaptation of many cultures. As a third of the year is provided by rainfall on average, the plants are supplied with a sufficient amount of water. A large number of sunny hours during the year leads to the proper development and normal performance of biological processes of plants. Plant species within the buffer zone show resistance to extremes, especially frost. Most species are also resistant to dry periods and high temperatures.

When measuring the area of the buffer zone using Google Earth, an area of 63,608.59 m² was obtained. Field observation revealed the presence of seven woody plant species and three shrub species. The most common plant species is *Populus alba* L. It covers

approximately 70% of the territory. In terms of number, the next is *Picea abies* L. It occupies only about 8% of the territory of the buffer zone (Figure 2).

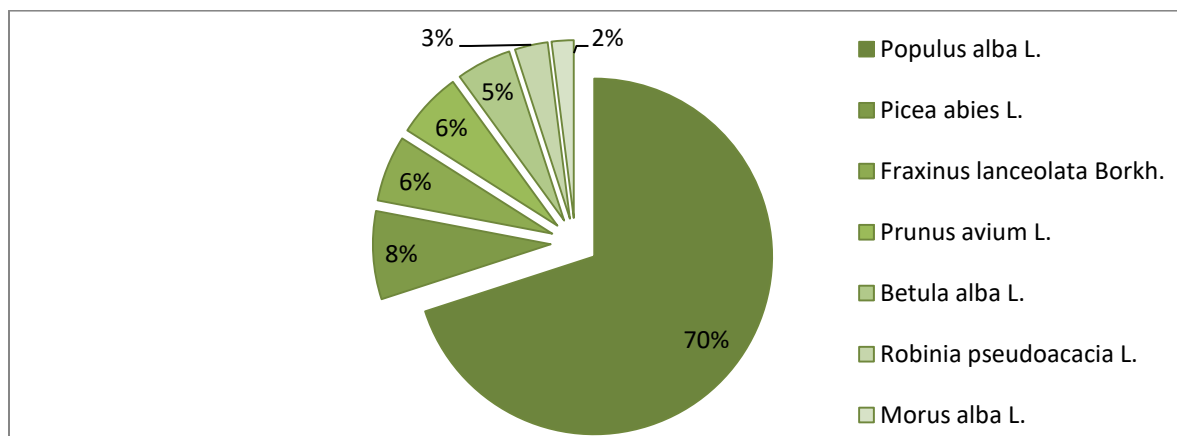


Figure 2. The share of woody plant species within the buffer zone

As it is the most represented here, the individuals of the *Populus alba* L. are very variable (Figure 3). Most individuals show the same characteristics, but there some trees are older than others. Taking average values into account, the height of poplar individuals within the buffer zone is 28 m, while the height of trunk to first branches is 2 m. Individuals that are in a denser plant (the distance between individuals is 2.5 m) have a much thinner trunk diameter of 0.4 m. A smaller percentage of individuals have a fairly wide trunk diameter of 1.3 m. The crown width of most individuals is 3.5 m, so in denser plantings they intertwine (Table 1). No type of damage is present, and these individuals receive the highest vitality rating (5). Also, undergrowth is present with most individuals.

Fast-growing *Populus* species are good for phytoremediation purposes because they have extensive root systems, high rates of water absorption and transpiration, resulting in rapid growth and high biomass. Poplars can be grown in a wide range of climatic conditions and unlike other genera, poplars can simultaneously be used for several different types of phytoremediation (Peuke & Rennenberg, 2005). As many studies show poplars as the best choice for a range of pollution (Haycock et al., 1996; Peuke & Rennenberg, 2005; Berlizov et al., 2008; Aghaalkhani et al., 2017; Yadav et al., 2010; Robinson et al., 2000), the analyzed buffer zone, i. e. the vegetation of the study area has a great potential in cleaning the polluted land and securing the arable land located near the buffer zone.



Figure 3. Variability of poplar individuals
(Source: authors, 2020)

It is clear that the vegetation is necessary to maintain the organic matter in the soil, which is needed for maintaining processes such as denitrification. Woody vegetation is also better at providing organic matter in the deeper subsoils, where it is needed for effective denitrification of groundwater (Haycock et al., 1996). Based on all the above, the species within the buffer zone must be adapted to the existing conditions of the location. Buffer zones are a subject of research by many authors and a number of research studies have reported positive effects of buffer zones (Haycock et al., 1996, Ebregt & De Greve, 2000, Norris, 1993, Parkyn, 2004, Osborne & Kovacic, 1993, Sahu & Gu, 2009). As in accordance to other authors they are a necessity in agriculture and nature preservation and a part of sustainable rural landscapes and urban landscapes.

Besides *Populus alba* L. as the dominant species in the buffer zone there are other species with different heights and habitus that contribute to the effectiveness of the buffer zone. Ash individuals, *Fraxinus lanceolata* Borkh. are located along the border of the buffer zone, to the left of the factory (towards Bač), intermittently planted with individuals of wild cherry, *Prunus avium* L. The average height of individuals of ash is 20 m, while the height of the trunk to the first branches is 2 m. The trees are surrounded by bushes of rosehip, *Rosa canina* L. and black elder, *Sambucus nigra* L. Trunk diameter of the individuals is 0.6 m, and the crowns are 7 m wide (Table 1). No type of damage is present on the trees, and the vitality rating of this species is the highest (5). *Prunus avium* L. within the buffer zone are 18 m high, and the height of the trunk to the first branches is 1.6 m. The trunk is surrounded by black elders and rosehips. The trunk diameter is the same as that of ash (0.6 m), while the crown width is 6 m (Table 1). Mechanical damage is not present, but phytopathological damage is evident on the leaves. Namely, there are necrotic spots on the leaf. The vitality rating of this species is therefore 4. Birch trees, *Betula alba* L. are found in this area right in front of the factory buildings. They are planted at a distance of 2.5 m. Their height is 15 m, the height of the first branches is 3.5 m, while the trunk diameter is 0.3 m. The crown is airy, 2.5 m wide (Table 1). Damages are not visible. The vitality score is high (5). *Robinia pseudoacacia* L. occurs in the area of the buffer zone as a wild, weedy species. It has the form of a bush, is 2 m tall and has the same crown width (2 m). The height of the trunk to the first branches is

only 15 cm, and the trunk diameter is 10 cm (Table 1). The vitality rating of this species is high (5), but acacia in this area can be considered a weed. Individuals of *Morus alba* L., have the smallest share in this area. The height of individuals is 9 m, while the height of the trunk to the first branches is 0.9 m. The crown is airy, sparse, and 3 m wide. Trunk diameter is 0.3 m (Table 1). Mechanical damage is not present, but necrotic spots appear on the leaves. That is why the individual's vitality rating is 4. Spruce, *Picea abies* L., as the only coniferous species in this area, was used to emphasize the entrance to the factory itself. It has a pronounced decorativeness, it is 20 m high, and the trunk up to the first branches is 0.5 m high. The chest diameter of the trunk is 0.4 m. The crown is 10 m wide (Table 1). In front of the spruce trees, following the road to the factory, there is a hedge of Wilson's honeysuckle, *Lonicera nitida* Wils. *Rosa canina* L. is very widespread in the area of the buffer zone, with an average height of about 1.8 m and a width of 3 m (Table 1). It is spread on the ground or leaning against trees. Fruits are present on individuals. It has a healthy appearance and the vitality rating of this species is the highest 5. Black elder bushes, *Sambucus nigra* L. are quite widespread in this area. They are on average 5 m high, and their width is about 4 m (Table 1). They occupy spaces around tree trunks or stand as independent shrubs. Wilson's honeysuckle, *Lonicera nitida* Wils. was, in this area, used as a hedge that emphasizes the entrance to the factory. However, it is neglected and needs pruning. It is 1.2 m high and spread to about 2 m in width (Table 1). To make it more decorative, it needs to be pruned, but the vitality of this species is quite high (5).

Table 1. Characteristics of the species within the Buffer zone
(Source: authors, 2020)

Species	Height (m)	Trunk height (m)	Trunk diameter (m)	Rotting of the trunk	Crown width (m)	Dried branches	Cut thicker branches	Dried crown top	Mechanical damage	Phytopat. damage	Vitality score
<i>Populus alba</i> L.	28	2	0,4	-	3,5	-	-	-	-	-	5
<i>Fraxinus lanceolata</i> Borkh.	20	2	0,6	-	7	-	-	-	-	-	5
<i>Prunus avium</i> L.	18	1,6	0,6	-	6	-	-	-	-	*	4
<i>Betula alba</i> L.	15	3,5	0,3	-	2,5	-	-	-	-	-	5
<i>Robinia pseudoacacia</i> L.	2	0,15	0,1	-	2	-	-	-	-	-	5
<i>Morus alba</i> L.	9	0,9	0,3	-	3	-	-	-	-	*	4
<i>Picea abies</i> L.	20	0,5	0,4	-	10	-	-	-	-	-	5
<i>Rosa canina</i> L.	1,8	-	-	-	3	-	-	-	-	-	5
<i>Sambucus nigra</i> L.	5	0,2	-	-	4	-	-	-	-	-	5
<i>Lonicera nitida</i> Wils.	1,2	-	-	-	2	-	-	-	-	-	5

*Present

The results show that the buffer zone individuals are well adapted to the current location conditions (Figure 4). The width of the buffer zone is sufficient as it provides a visual barrier from the factory that also reduces noise pollution. As tree species dominate, to

increase the effectiveness of the buffer zone in protecting the arable land from pollution (Rahim et al., 2014, Albe, 2003, Park & Yoo, 2004) stated that vegetation is also necessary in the lower stories. As there are some gaps in the buffer zone with the lack of lower vegetation, some further research is needed to provide information if these gaps make problems in the effectiveness of the buffer zone.



Figure 4. Individuals within buffer zone
(Source: authors, 2020)

CONCLUSION

The buffer zone is located in an area with a temperate-continental climate that corresponds to the development of the present plant species. Also, for the proper growth and development of existing plant species, the soil on which they develop is suitable. There are six deciduous woody species (*Populus alba* L., *Fraxinus lanceolata* Borkh., *Prunus avium* L., *Betula alba* L., *Robinia pseudoacacia* L., *Morus alba* L.), one coniferous woody species (*Picea abies* L.) and three shrub species (*Rosa canina* L., *Sambucus nigra* L., *Lonicera nitida* Wils.). The vitality of all species within this area is at a very high level. In this area, there is undergrowth, especially poplar, which needs to be controlled. The space should be maintained more regularly so that weed plants do not develop. Wilson's honeysuckle hedges also need regular pruning. Since the area itself is a factory complex, the maintenance of the buffer zone is of great importance because this type of greenery helps control the quality of water, air and soil, and solves many environmental problems. The proximity of the Special Nature Reserve "Karadorđevo" and the Danube-Tisa-Danube canal is an additional reason for maintaining this area. The buffer zone around Sugar Refining Mill in Bač is a great example of how landscape planners, landscape architects and nature conservationists can have a big role in protection of agricultural land around industrial facilities using buffer zones as solutions based on nature that are a key to sustainability.

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SURGICAL MANAGEMENT OF AN INFECTED WING WOUND IN A CAPERCAILLIE (*Tetrao urogallus*)- A CASE REPORT

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SUMMARY

*Capercaillie (*Tetrao urogallus*) is a strictly protected wild species and the total breeding population is estimated to be 50-100 singing males in the Republic of Croatia. In Gorski kotar, Croatia there is a "Central European International Center for the Growing of Capercaillie in Gorski kotar (Republic of Croatia)" whose role is the revitalization of capercaillie and reinhabiting these birds to the Gorski kotar, their natural habitat. We treated a two-month-old bird from that center. The bird had an infected wound on the left wing and we performed extensive surgical treatment of the wound with debridement of complete scar tissue, cartilage and subchondral bone with skin reconstruction. As premedication, we used medetomidine, butorphanol, midazolam and ketamine as an intramuscular injection. Surgery was performed under general inhalation anaesthesia with sevoflurane. Healing of the wound and soft tissue postoperatively was relatively fast and this surgical procedure proved extremely important for the preservation and survival of the remaining small number of capercaillies found in the Republic of Croatia.*

Keywords: *Tetrao urogallus, capercaillie, wound, infection, surgery, anaesthesia*

INTRODUCTION

Capercaillie (*Tetrao urogallus*) is Croatia's largest member of the forest grouses (*Tetraoninae*) (BAGARIĆ, 2016) and belongs to the order Galliformes (RADOVIĆ et al., 2003). Capercaillie is a strictly protected wild species in the Republic of Croatia (ANONYMUS, 2013). According to the currently valid classification, capercaillie is a qualifying species for two areas of conservation important for birds. These two areas are Gorski kotar and northern Lika (HR1000019) and also Velebit (HR1000022) (ANONYMUS, 2019). The distribution represents the uninterrupted area in the northern forests of Eurasia while areas in central Europe are extremely fragmented with many small, separated mountain populations fragmented into little areas (BRAUNISCH et al., 2008). The capercaillie is a species that lives predominantly in old and natural forests in Europe, in the spruce-beech-fir and spruce vegetation belt (KOLETIĆ, 2018) and nests at altitudes of over 900 meters, so the tracking is very difficult (BAGARIĆ, 2016). Adult birds mainly feed on plant food, while most of the animal prey is caught by accident with plant food (BAGARIĆ, 2016). The average length of capercaillie in our region is 65-70 cm (from the tip of the beak to the root of the tail). The average body weight is 7 kg, while females are smaller and

lighter. The color of the male feathers is black with a white spot on the wings. He also has a beard with longer feathers on his chin, and a red rose above his eyes. Females are brown, like earth (JANICKI et al., 2007). Capercaillie is one of the symbols of the mountain part of Croatia. The number of them in Croatia is decreasing, as elsewhere in Europe. The reasons for the negative trend are numerous and still not totally understood (BAGARIĆ, 2016). The backbone of international protection of capercaillie is the Directive on the conservation of wild birds (2009/147/EZ). According to the criteria of the IUCN (International Union for Conservation of Nature), the species is currently classified as a species of least concern (LC). It's estimated that the population is declining owing to habitat destruction and alteration (BirdLife International, 2015). In the Red Book of Threatened Birds of Croatia (TUTIŠ et al., 2013), the capercaillie is classified as an endangered (EN) species, due to the declining trend of the population. The total breeding population is estimated to be 50-100 singing males. The population in Croatia is weakly investigated and it is not possible to define with certainty the causes of endangerment. Probably the most important causes of endangerment are forest management which leads to a reduction of habitat quality, poaching and an increase in numbers of game, especially wild boar, due to feeding. As a ground-nesting bird, this species is very sensitive to disturbance, so the increase in the number of the wild game has a negative impact on the success of their nesting. An increase in the number of wild boars also leads to increased predation on eggs and young capercaillie birds. With the opening of construction or forest roads, birds are additionally disturbed, as well as by increasing tourism and recreational activities in mountain areas (TUTIŠ et al., 2013). The bird from this case report came from the "Capercaillie association", the breeding center located in Prezid, in Gorski kotar. The main goal of the "Central European International Centre for the Growing of Capercaillie in Gorski kotar (Republic of Croatia)" is the revitalization of capercaillie and reinhabiting these birds to the Gorski kotar, their natural habitat. The Association cooperates with the Faculty of Veterinary medicine since 2018.



Picture 1. A large (6000 m²) aviary in "Central European International Centre for the Growing of Capercaillie in Gorski Kotar (Republic of Croatia)" (<https://capercaillie.eu/en-wildlife-photography.html>)

This is a work of great importance because capercaillie is a protected species and all new knowledge can help to preserve this endangered species. That's why we wrote a case report, surgical management of an infected wing wound in capercaillie (*Tetrao urogallus*) from August 2022. As far as we know, there are no studies that report an infected wing wound in

capercaillie (*Tetrao urogallus*) or surgical management of this condition, or anaesthesia protocol for this species.

CASE PRESENTATION

Capercaillie (*Tetrao urogallus*) named Eva was admitted to the Clinic for surgery, orthopaedics and ophthalmology at the Faculty of Veterinary Medicine at the University of Zagreb for the treatment of the wound tissue formation on the left wing.



Picture 2. The wound on the left wing

The patient was two-month-old female capercaillie. The bird was bitten by a fox and the bleeding on the bite site was severe. The local veterinarian decided to cut off part of the wing which was bleeding. The animal protector said that he tied the wing proximally to the bleeding site and cut part of the wing in the joint. The patient was presented with noticed suspicious growth of tissue on the left wing. The patient was making painful noises and she was restless. The animal protector also noted that the bird was pecking at the wound and that the other birds were also trying to peck at it. As he stated, the bird was healthy before the incident. In the clinical examination, the bird's heart rate was 190 beats per minute and her temperature was 39,4°C. The bird was calm and she was often making noises. The patient was aware of her surroundings, body condition was within the physiological value, the body weight was 0,8 kg. The loss of feathers was visible in the area of the end of the left radius and ulna, while the feathers on the rest of the body were neat and undamaged. In that area was visible a mass, with a hard consistency, without feathers, the size and shape of a cherry. The bird was offered an apple and water and she was kept overnight at the Clinic for further diagnostics. The finding of the X-ray was that on the distal part of the wing there is a visible lack of bone macrostructure from the metacarpal bones more distally. In the area of the end of the radius and ulna, a mass of heterogeneous intensity with a round shadow was visible. Bone structures were also visualized in the area of swelling. A diagnosis, tumor in regionem metacarpalis et phalangis allae sinistrae, was made.



Picture 3. Radiographs of the left wing

The treatment plan was to do surgery under general anaesthesia in order to remove the whole mass. Premedication was given in the form of an intramuscular injection in the pectoral muscle. Premedication consisted of the following drugs and doses: medetomidine (dose 20 µg/kg; α_2 -adrenoceptor agonist), butorphanol (1 mg/kg; agonist-antagonist opioid analgetic), midazolam (6 mg/kg; benzodiazepine) and ketamine (10 mg/kg; dissociative anesthetic).



Picture 4. Measurement of heartbeats with a Doppler device in anesthetized, intubated patient



Picture 5. Drugs used in premedication

The intravenous cannula was placed on the healthy wing. The bird was intubated and during the surgery she was breathing spontaneously. Sevoflurane was used (as an inhalation anesthetic) in a mixture with air and oxygen. The patient was receiving intravenous fluid therapy, Hartmann's solution 10 ml/kg/h. Intraoperatively the bird was initially stable and later bradycardic.

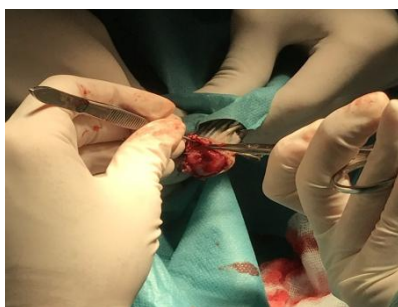


Picture 6. Bird immediately after the surgery

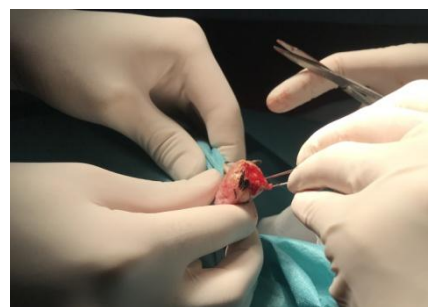


Picture 7. Bird after fully waking up from anaesthesia

At the end of the surgery the patient got meloxicam intramuscularly (dose 1 mg/kg; nonsteroidal anti-inflammatory drug), enrofloxacin subcutaneously (dose 15 mg/kg; fluoroquinolone antibiotic) and atipamezole intramuscularly (same volume as α 2-adrenoceptor agonist; α 2- adrenergic receptor antagonist). The bird was apathetic in the cage after waking up. A few hours later, she started eating, drinking and defecating. In the afternoon she was given *Duphalyte*[™] subcutaneously (as additional fluid therapy). In surgery, radical debridement was performed, with debridement of complete scar tissue, cartilage and subchondral bone, followed by skin reconstruction, *Byosin*[™] USP 3-0.



Picture 8. Surgery on the capercaillie's left wing



Picture 9. Radical debridement and reconstruction

The bird was released from the Clinic the same day and she was given home therapy, consisted of enrofloxacin (*Enroxil*[™] 5%, 2 mL per 1 L of water through the whole day, perorally, during five days) and meloxicam oral suspension (*Meloxoral*[™] (1,5 mg/mL), 1 drop in the beak, during five days).

DISCUSSION AND CONCLUSION

The wound was caused by improper treatment of the initial bite wound. Tissue was surgically removed up to the level of the subchondral bone in order to enable and fasten tissue healing. Possible complications include infection of the surgical wound, dehiscence of sutures, delayed healing due to holding conditions and contamination of the wound by external influence. The prognosis was questionable due to our lack of experience in such procedures. A few months after the wound healed without any complication.

This was one of three similar cases where we have performed on capercaillie in the last few months. The anaesthesia and post-operative recovery went without complications.

This is, to our knowledge, the first case of surgical management of an infected wing wound and specific anaesthesia protocol reported in this species.

CONCLUSION

1. Capercaillie anaesthesia is complex and proved to be successful with intramuscular premedication (medetomidine, butorphanol, midazolam and ketamine) and inhalation anaesthetic (sevoflurane).
2. Healing of the wound monitored postoperatively was relatively fast in comparison to other animal species.
3. The patient was discharged the same afternoon in order to return to natural habitat as soon as possible.
4. This is one of our three similar cases of surgery on capercaillie successfully performed, which shows a favorable prognosis for cases with similar pathology.
5. This surgical treatment proved extremely important point for the preservation and survival of remaining capercaillies found in the Republic of Croatia.

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EFFECT OF ADIPOSE TISSUE AUTOLOGOUS GRAFT ON HEALING OF HENS' DIAPHYSIS CRITICAL DEFECT

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SUMMARY

Bone is a tissue that has a great healing capability with the equivalent tissue without forming a cicatrix. However, when it comes to extensive fractures the bone cannot heal on its own. Therefore, in those cases different types of assisted healing procedures are taken into consideration. In this research it was studied how will the graft from the subcutaneous and abdominal adipose tissue effect on the critical bone defect on diaphyseal bone in hens. For that reason, 30 hens were used, hybrids of light breeds that were exuded from the production. The hens were divided into three groups (A, B, C). One group (A) was a control group whereas the other groups were experimental. Every group contained 10 individuals. After doing a critical bone defect on ulna, in 10 individuals a subcutaneous adipose tissue implantation was performed (group B) and in the next 10 individuals an abdominal adipose tissue implantation was performed (group C). In the 10 remaining hens the critical bone defect was left empty. The effect of the autologous native adipose tissue graft was histologically and radiographically studied in these groups. In order to monitor the healing process, control examinations were done immediately after the surgical procedure, after 3.5 weeks after the procedure and 7 weeks after the procedure. Histological findings showed that there were statistically significant differences between the type of the newly formed bone, formed callus, and in the bone bridging between the control group (A) and the subcutaneous group (B) as well as between the control group and the abdominal group (C) in hens observed after 3.5 and 7 weeks. Also, when observed collectively after 3.5 and 7 weeks, statistically significant findings were found between the control (A) and subcutaneous (B) group as well as between the control (A) and abdominal (C) group. Histological findings were confirmed by radiographic findings. X-ray images showed that the adipose tissue (especially the abdominal one) improved the healing process of the critical defect. Findings in the control group (such as poorly formed callus and an empty bone crack) were not considered as a part of the healing process.

Key words: bone defect, adipose tissue, healing, hens

INTRODUCTION

Large bone defects and incurable fractures have always been challenging to treat. There are several possibilities, including autologous bone grafts, cadaveric bone grafts, free tissue transfer and allotransplantation (Bohnenblust et al., 2009). Stem cells can also be used to help with tissue regeneration as the newest method of treatment. Adipose-derived cells (ASC) have stood out the most, having the ability to renew themselves, commit and multiply cell lineages (Gimble et al., 2007). They can also be easily singled out with minimal damage (Fraser et al., 2006, Fitzsimmons et al., 2018). It has been confirmed that the ASCs are a source of multipotent cells with osteogenic differentiation potential *in vivo* and *in vitro* (Wang et al., 2009), although the *in vivo* usage in order to repair bone fractures has not been largely evaluated. Taking into consideration the easy access to adipose tissue through resection or liposuction and its high number of stem cells, its use in bone engineering ought to be an exquisite method. Specifically in bird bone engineering, avian bone is structurally and biologically different so there are still no distinct options available. Since bone fractures are one of the most common conditions found in avian patients, it is highly desirable and important to keep investigating the possible treatment options. The aim of this research was to study the effect of stimulated healing in a critical ulna defect in exuded hybrids of light breeds in hens.

MATERIALS AND METHODS

In this research 30 laying hens were used, 14 months old hybrids of light breeds, that were exuded from the production. The hens have been transported to the experimental rooms of the Department of poultry diseases with clinic of Veterinary Faculty in Zagreb, Croatia. The hens were marked with leg rings from 1 to 30 and their incoming weight was measured with the laboratory electronic beam scale (± 0.1 g). Their weight ranged from 1.344 g and 2.036 g and the average weight was 1.742 g per hen. The hens were fed with the same feed mixture for poultry (120g hen/day) with which they have been fed before coming to the Faculty. The water was available *ad libitum* through the nipple system.

The hens in this research were divided into 3 groups (1 control group and 2 experimental groups). Control group (A) consisted of 10 individuals, experimental group (B) with implantation of adipose tissue from the subcutis consisted of 10 individuals and experimental group (C) with implantation of adipose tissue from the abdomen consisted also of 10 individuals.

Five days after their arrival, a surgical procedure was done. Enrofloxacin (Vetoflok 5%, Veterina d.o.o., Croatia) was given to the hens as a part of the preoperative and postoperative care (10 mg/kg b.m.). Premedication consisted of Diazepam 0.5 mg/kg b.m. (Apaurin, Krka, Slovenia) and Ketamine hydrochloride 20 mg/kg b.m. (Narketan, Vetoquinol, Switzerland). The surgical procedure consisted of 2 steps. The first step was to explant the adipose tissue graft. In order to do that, the feathers on the left side of the abdomen were removed and the operative field was aseptically prepared in 20 hens (10 from group B and 10 from group C).

A 3 cm cut was made in order to exploit adipose tissue graft in the size of 10 x15 mm. Extracted adipose tissue was put in the sterile saline solution. The wound was closed layer by layer with Dexon 3-0 (Dexon S, Medtronic, USA) and Seralon 4-0 (Seralon , Serag-Wiessner GmbH & Co.)

The second part of the procedure was to perform a cut 3 cm long in the area of the dorsal middle third of the right elbow bone in all 30 hens. Orthopedic oscillating saw Aesculap was used to perform a 15 mm long resection of the bone cylinder of the ulna. The bone defect in hens from experimental group B was implanted with the previously taken autologous subcutaneous adipose tissue graft, whilst the bone defect in hens from experimental group C was implanted with the autologous abdominal adipose tissue graft. Implantation of adipose tissue graft was not performed on the critical bone defect in hens from the control group (group A). The wound was closed in layers with Dexon 4-0 (Dexon S, Medtronic, USA) and Seralon 4-0 (Seralon, Serag-Wiessner GmbH & Co.). While observing the healing process of the diaphysis, fixation bandages were used to fixate the wings of operated hens. In order to protect the wound from secondary contamination, an elastic transparent bandage was used on the right wing of every individual. Consequently, a continuous fixation was given to the forearm which was strong enough to maintain immobilization without any further traumatization of the hens.

Three and a half weeks after the ulnar resection, the right forearm was removed in 14 out of 29 individuals (one individual died from a gastrointestinal disorder). Out of those 14 individuals, 4 were from the control group, 5 were from the experimental group B and finally, 5 were from the experimental group C. The removed right ulna was preserved in 10% formalin solution because they were further processed for the preparation of histological samples. Seven weeks after the ulnar resection, the right forearm was removed in the same way in the remaining 15 hens (5 from each group) for the purpose of making histological samples and X-ray images.

To monitor the healing process of the critical bone defect in the ulna, X-ray images were taken. The images were taken with an X-ray device (Siemens, Germany) with the exposition of 40 kV and 6mAs and they were interpreted with a digital reader Agfa CR30-X. Evaluation of the radiographic changes in the critical ulna defect was performed immediately after the surgical procedure in all 30 hens. The second control was performed 3,5 weeks after the surgical procedure in 29 hens. The third control was performed 7 weeks after the surgical procedure in 15 hens (5 from each group). The images were taken in a sagittal plane.

In order to get histological samples, all bone samples were first cut with a dental circular saw under an angle of 45 degrees on the longitudinal axis of the bone. From there on they were put in a 10% formaldehyde solution. The fixation of the samples was done at the room temperature and the ratio of the sample and the fixative was 1:20 through 48 hours. After that, the samples were demineralized for 14 days in a mixture of 15% aqueous formic acid solution and 4% chloride acid solution (SUVARNA et al., 2013.) The given samples were embedded in paraffin blocks which have been cut 3mm thick by microtome (Mikrom). The histological samples were fixated on microscope slides for further proceedings of hematoxylin – eosin (HE) staining and van Gieson staining for connective tissue. These samples were analyzed under the microscope searching for signs of new bone formation.

Also, the type of bone was determined (woven /primary, lamellar bone or both). Furthermore, type of the bone callus was determined (bone, fibrocartilage callus or mixture). Bone bridging of the defect was marked as present (+) or not present (-). These parameters were observed in order to support the analysis of X-ray findings.

Statistical analysis was performed in STATISTICA v.13.5. (Statistica, Inc. 2018.) program. Differences between groups for the variables' presence of the bridging and type of callus were analyzed using the Freeman-Halton extension of Fischer's exact test. This test is used as an alternative to the χ^2 test in cases where the number of measurements is low and when the contingency table is greater than 2x2. All differences were observed at the level of statistical significance $P < 0.05$.

RESULTS AND DISCUSSION

In this research, the effect of a free graft of subcutaneous and abdominal adipose tissue on the healing process of a critical-size ulnar defect in hens was investigated. Healing was defined as bridging of the critical ulnar defect or filling the defect with a callus (not necessarily the complete defect), whereas the critical size ulnar defect was defined as the smallest intra-bone wound that does not heal during the lifetime of the organism. Bone bridging and callus formation were observed in experimental groups (B-subcutaneous group and C-abdominal experimental group) of hens at 3,5 and 7 weeks after the surgical procedure. In group B (subcutaneous group), 8/10 hens have bridged the defect (although not completely) in a period of 7 weeks. In group C (abdominal group), 10/10 hens have bridged the defect (although not completely) in a period of 7 weeks. Bone bridging and callus formation did not occur in group A (the control group). Histological analysis, as well as the X-ray imaging, showed that most of the newly formed bone tissue, created on the edges of the critical size ulnar defects which partially filled the defect, was synthesized under the influence of the autologous graft of the adipose tissue from the abdominal region (10/10). A certain degree of bone tissue formation and bridging of the critical size ulnar defect was also seen in the group that had the subcutaneous autologous graft implemented, but with a weaker outcome (8/10) (Figure 1.). Histologically, the implanted adipose tissue graft visible showed good vitality even after 3,5 weeks, meaning that dying off of the adipocytes was not present. Statistically significant differences ($p < 0,05$) were recorded when observing the type of the newly formed bone, callus formation and the presence of bone bridging. The previously mentioned statistically significant differences were noted between the control group (A) and subcutaneous group (B), as well as between the control group (A) and the abdominal group (C) observed after 3.5 and 7 weeks. Also, when observed collectively after 3.5 and 7 weeks, statistically significant findings were found between the control (A) and subcutaneous (B) group as well as between the control (A) and abdominal (C) group. The results have shown that adipose tissue, especially from the abdominal region, was a successful factor in bridging of the artificially made critical size ulnar defect since it is rich in blood vessels and has less fibrous encapsulation compared to the subcutaneous one. A bigger amount of newly formed bone tissue can be explained by its osteoinductive features because of the growth factors that

significantly expand its regenerative potential. During this research, complete regeneration has not occurred in either of the groups in the given time of seven weeks, which implies that further investigation of adipose tissue grafts is necessary, especially during a longer period of time (over 8 weeks). Also, it is left to explore the biomechanical properties of the newly formed bone tissue which healed with the help of autologous adipose tissue graft. A new hypothesis can also be set for further researches – exploring significantly more possibilities of stimulated bone formation *in vivo* which could lead to a connection with clinical studies conducted on people and could possibly make ASC useful for future tissue engineering in clinical conditions.

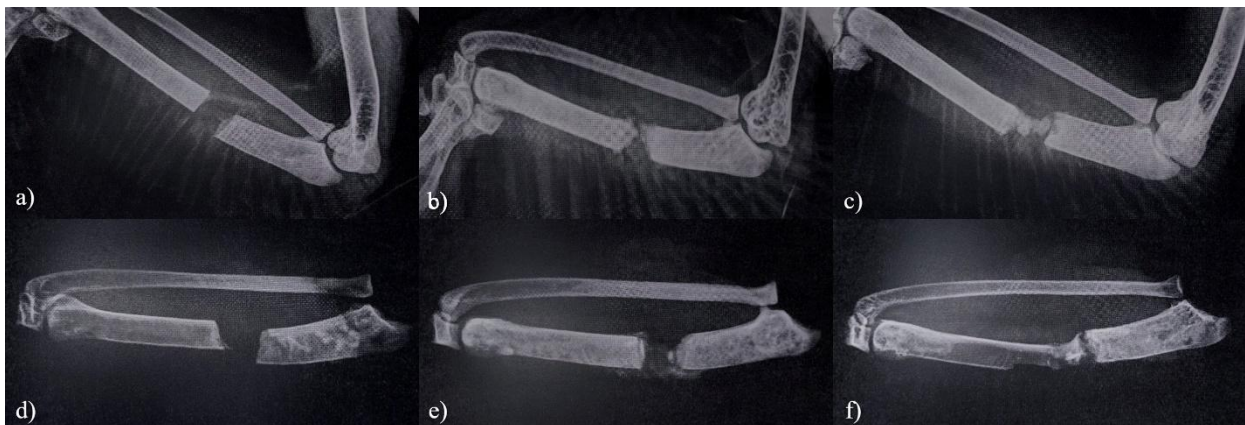


Figure 1. X-ray images of the healing process. Pictures a) to c) show the process of healing 3.5 weeks after the surgical procedure (a- control group, b- subcutaneous experimental group, c- abdominal experimental group). Pictures d) to f) show the process of healing 7 weeks after the surgical procedure (d- control group, e- subcutaneous group, f- abdominal experimental group).

CONCLUSIONS

This research shows that an autologous adipose tissue graft has a potential role in the healing process of the critical bone defect. X-ray findings 7 weeks after the surgical procedure show that the abdominal adipose tissue graft has an effect on partial bone bridging. Critical bone defects do not heal on their own, which is why the use of the autologous graft is a confirmation of its effect on the healing process. In comparison to the use of stem cells, adipose tissue advantages are its' availability and amount, the surgical procedure is less invasive and it contains more stem cells than the bone marrow. Further researches should be aimed at the use of biomarkers, which would help in the better understanding of biological processes in the healing of the critical bone defect with adipose tissue. Also, biomechanical characteristics of the healed critical defect should be investigated.

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PERCEIVED STRESS AMONG THE STUDENTS OF THE FACULTY OF AGRICULTURE NOVI SAD

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SUMMARY

Due to the impact that it has on the performance of individuals, stress as a phenomenon is receiving increasingly more attention from researchers. This study aims to measure the stress level of students attending the University of Novi Sad, with a particular focus on the students of the Faculty of Agriculture. A revised version of the Perceived Stress Scale–PSS was used to measure the perceived stress level. The results indicate that most of the respondents (64.36%) were experiencing moderate levels of stress. A somewhat lower level of perceived stress was recorded for the students of the Faculty of Agriculture (20.11) when compared to that of students attending other faculties at the same university (22.94). This could, at least in part, be ascribed to the differences in class organization at various faculties at a time when functioning was impeded by the implementation of counter-epidemic measures.

Key words: stress, perceived stress scale, students, COVID-19

INTRODUCTION

Stress is a state of imbalance in the human body triggered by the differences between the requirements of the environment and an individual's capacity to respond to those requirements (Subramaniam, 2017). It manifests in the form of various physiological, emotional, and behavioral effects (Lazarus, 1984). Due to the impact of stress on the health and performance of employees, and consequently on the performance of companies, stress has often been studied in the field of human resource management (Avey et al., 2009; Voci et al., 2016). To date, no consensus has been achieved among researchers studying stress regarding its impact on the performance of employees, i.e. whether it has both a positive/motivating and negative effect (depending on the quality and quantity of the stressor which the individual is exposed to and how they react to it), or if the effects of stress are always negative (Welford, 2007; Westman and Eden, 2007; Gino, 2016; Hargrove et al., 2016). Even though the so-called Yerkes-Dodson Law has for decades been used to justify managerial practices which cause employees stress (Tiegen, 1994), studies have shown that there is no valid scientific evidence that stress affects the performance of employees along an inverted U-shaped curve (Muse et al., 2009; Corbett, 2015).

The student population is usually included in this kind of research as well, since it has been proven that changes in one's lifestyle, which occur at the onset of the tertiary level of

education, result in an increase in the stress levels of students (Jovanović and Gavrilov-Jerković, 2015; Kwok and NG, 2016; Silva et al., 2018). Finding adequate strategies to manage stress in this population has a considerable social impact since stress, in addition the impact it has on their current performance, if it becomes chronic, can have an impact on their future performances as well (Gazzaz et al., 2018). Therefore, both universities and prospective employers should be interested in this particular issue.

One of the key research questions which continue to be current is how to measure stress levels. Measuring stress levels is a precondition for taking action aimed at the elimination of those stressors which lead to a decrease in the performance of individuals (Subramaniam, 2017). With that aim in mind, various means of measuring stress levels are applied. One of the most frequently used instruments for this purpose is the PSS-Perceived Stress Scale, which was also used in this study and which will be discussed in more detail later. In addition to the PSS-scale for measuring stress, frequently used scales include the PSQ-Perceived Stress Questionnaire (Levenstein et al., 1993) and the DASS21-The Depression Anxiety and Stress scale (Lovibond et al., 1995, as cited in Jovanović et al., 2011).

The aim of this study is to determine the perceived stress level among the population of students of the Agricultural Faculty and compare it to that of students from the other faculties of the University of Novi Sad (UNS), and to use the results obtained to provide recommendations for the faculty management. It is important to mention that the perceived level of stress among students was analyzed at a time when they were most likely under the considerable influence of the global COVID-19 pandemic. This led to online education, which had previously not been the practice. As a result, we can assume that the perceived stress level was higher than the one that would have been recorded under more regular circumstances. However, since virtual classes were a novel experience for all the students of the UNS, and the counter-pandemic measures were the same for everyone, this fact should not have a significant impact on the determined differences among the studied groups, and thus the derived conclusions.

THE METHOD

To realize the set aims, a survey research method was selected. The measuring instrument was the PSS – Perceived Stress Scale, which, as pointed out in the introduction, is one of the most frequently used instruments for this very purpose. It was developed by Cohen and associates to determine the stress level of certain life situations (Cohen et al., 1983; Bourion-Bédès et al., 2021). The original version of the scale consists of 14 questions, but today a revised PSS scale of 10 questions is seeing increased use in research¹ (Bourion-Bédès et al., 2021). It was used as the measuring instrument in this study. Its reliability was confirmed by Cronbach's Alpha coefficient (0.842).

The anonymous study included 275 students, who voluntarily agreed to participate, of which 103 were students of the Faculty of Agriculture in Novi Sad. The study was carried out

¹<https://www.cmu.edu/dietrich/psychology/stress-immunity-disease-lab/scales/revised-pss-request-reply-for-all-requests.pdf>

by means of a so-called online survey, where by the questions were sent to known addresses, along with a request to further distribute them to other students of the UNS. Even though this survey method and the samples obtained in this manner have certain disadvantages (Evans and Mathur, 2005), due to low costs and ease of realization, this type of survey is increasingly more prevalent (Sax et al., 2003). The study was carried out from December 31, 2020 to January 8, 2021. This study period was deliberately chosen with the aim of analyzing the perceived stress levels among the student population over a holiday period (when students are usually more relaxed), but which precedes their exam term. Responses to the questions were evaluated on a Likert-type scale ranging from 0 to 4 (never / almost never / sometimes / often / very often), while in the case of positively formed questions the scoring was on a scale of 4 to 0, with the same intervals. A total score was calculated for each individual (min 0, max 40). The greater the value of the score, the higher the perceived stress level (Cohen et al., 1983). Based on the total score, the respondents were classified into three groups based on perceived stress levels (Cohen et al., 1983; AlAteeq et al., 2020): a) a low level (0-13); b) a moderate level (14-26), and c) a high level (27-40). Basic methods of descriptive statistics were used to process the data. Differences in the perceived stress levels of the students attending various faculties were determined using an independent samples t-test.

THE RESULTS AND DISCUSSION

The sample included 275 students of the UNS. Considering the aim of the study, the focus was on the students of the Faculty of Agriculture, and so more than one-third of the surveyed respondents were in fact attending this faculty (Table 1). A greater number of female students in the sample was the outcome of a greater readiness on the part of the female students to take part in the survey (Smith, 2008). When analyzing the results, the structure of the sample needs to be taken into account, since it could have an impact on the results.

Table 1: Sample structure (N=275)

Demographic characteristics		Number of respondents	Participation in %
Gender	Male	82	29.8
	Female	193	70.2
Place of residence	Rural	83	30.2
	Urban	192	69.8
Faculty	Faculty of Agriculture	103	37.5
	Other faculties	172	62.5
Level of study	Undergraduate studies	221	80.4
	Other levels of study	54	19.6

Source: Author's analysis of the survey data

The average total score calculated on the PSS scale for all the respondents was 21.88, which is a moderate level of stress. These values are close to those obtained by other authors who measured stress levels by means of the PSS scale. A similar result (20.43) was obtained by Kostić and associates when measuring the experienced stress level of the students attending the University of Niš (Kostić et al., 2021). On average, Portuguese (Silva et al., 2018), French (Bourion-Bédès, 2021), and British students (Shaw et al., 2017) were all exposed to moderate levels of stress, and their average values on the PSS scale were 21.9, 19.2 and 17.9, respectively.

The data shown in Table 2 indicate that almost two-thirds of the respondents in our study are experiencing a moderate level of stress (64.36%).

Table 2: Share of different stress levels (N=275)

PSS score	Number of respondents	Participation in %
A low level of stress (0-13)	32	11.64
A moderate level of stress (14-26)	177	64.36
A high level of stress (27-40)	66	24.00
Total	275	100.00

Source: Author's analysis of the survey data

However, it is somewhat worrisome that almost one-quarter of the surveyed students (24%) had experienced a high level of stress in the month preceding the survey, while only a few of the students were experiencing a low level of stress (11.64%). Similar results were obtained for perceived stress levels of students in Saudi Arabia, where it was determined that 55% of the respondents were experiencing moderate, 30.2% high, and only 14.8% a low level of stress (AlAteeq et al., 2020).

Table 3 shows the distribution of the responses to the individual questions from the PSS scale. As indicated by the presented data, more than one half of the respondents (56.7%) felt nervous and stressed 'often' or 'very often' over the past month. A considerable percentage of the respondents, when answering other negatively formed questions regarding their feelings, stated that over the past month they had had such negative experiences either 'often' or 'very often'. Relatively few of the surveyed students responded with 'often' or 'very often' to questions from the PSS scale which refer to positive emotions.

Table 3: PSS scale answer distribution, % (N=275)

In the last month, how often have you ...	Never	Almost never	Sometimes	Often	Very often
... been upset because of something that happened unexpectedly?	4.0	12.7	45.5	25.8	12.0
... felt that you were unable to control the important things in your life?	7.6	14.9	38.9	25.1	13.5

... felt nervous and “stressed”?	2.2	6.9	34.2	29.1	27.6
... felt that things were going your way?	14.5	24.0	40.7	16.7	4.0
... found that you could not cope with all the things you had to do?	6.9	21.1	47.3	18.2	6.5
... felt confident about your ability to handle your personal problems?	6.2	21.1	35.3	25.8	11.6
... been able to control irritations in your life?	6.5	21.1	33.8	29.1	9.5
... felt that you were on top of things?	5.1	16.0	40.4	28.7	9.8
... been angered because of things that were outside of your control?	7.6	12.4	33.1	27.6	19.3
... felt difficulties were piling up so high that you could not overcome them?	14.2	18.2	32	19.3	16.4

Source: Author’s analysis of the survey data

Considering that this is a young population, and that stress has been proven to impact the health of an individual (Pascoe et al., 2020), it is important to deal with the causes of stress and try to find a solution which could at least in part mitigate this problem which is of general social importance. Since no previous research exists based on which we could make comparisons or draw conclusions, we can only assume that, among other things, the SARS-CoV-2 pandemic did have a certain impact on the perceived stress levels both among our students and those in other countries (Bourion-Bédès et al., 2021). Proof of this can be found in a comment given by one of the respondents from the Faculty of Medicine:

“Actually, the only stress I really experienced had to do with the corona virus. And I am not even trying to follow the news and information about it, I avoid it, but it just keeps on coming. There is too much information which is often contradictory, it is brainwashing me, it’s like there are no normal things to talk about anymore; life is pointless and rather difficult, not just because of the virus-related illness, but because of the psychological stress it causes. At this point I would feel more comfortable dying of anything than having to put up with this stress for another year ...”

(female student, 22 years of age, author’s translation).

It is a fact that the pandemic has caused numerous changes in how all spheres of society function, including institutions of higher education, as well as changes in how study programs are organized. As a result of the shift to distance learning, the interaction between the students, professors, and their teaching assistants in the classroom has disappeared. As it is something they were used to, this led to a certain level of stress. Confirmation can be found in a comment made by one of the students of the Faculty of Agriculture:

“...the coronavirus has caused considerable upheaval in my life, not as an illness, but as a considerable difficulty in the realization of what had to now been everyday responsibilities, ranging from studying for an exam, taking an exam, then online teaching is not at all good, simply put, it is necessary to experience interaction with a professor or teaching assistant in the lecture hall.”

(a student of the Faculty of Agriculture, 23 years of age, author's translation)

In accordance with the set aim, the perceived stress level of the respondents from the Faculty of Agriculture was compared to that of the students from other faculties of the UNS. These findings could be of some importance even under regular circumstances and are especially important when viewing the level of adaptability to these extraordinary circumstances not only on the part of the students, but the faculties as well.

In the studied sample, a statistically significant lower level of stress was noted for the respondents from the Faculty of Agriculture than that of the students from other faculties of the UNS (Table 4).

Table 4. Differences in the average stress level between students of the Faculty of Agriculture and other UNS students (t-test results)

Groups	Means	Std. deviation	p	Eta squared
Faculty of Agriculture	20.11	5.81	0.000	0.05
Other faculties	22.94	7.23		

Source: Author's analysis of the survey data

This indicates that not all the faculties responded in a timely fashion or in the best possible way to the changes caused by the pandemic, which can certainly be one of the reasons why the perceived stress levels determined for certain faculties differ. Confirmation of this fact can be found in the critiques made by respondents from some of the faculties of the UNS regarding the decisions of their alma maters:

"This month has been a disaster, I was in isolation due to the coronavirus, the dates for my midterms kept changing, and finally ended up being one right after the other; almost none of my professors and assistants met the students half way, it was all just a lot of talk about how easy it would all be. Public transport was almost completely cancelled; I barely made it back home. I am hoping that the situation will improve, or that there will at least be some understanding for the students, because this can really not go on..."

(male student, 20 years of age, author's translation)

"...what contributed to the rise in stress levels quite significantly was all the added piling up of exams which were all of a sudden just thrown into the schedule once the state of emergency was called off, with added limitations on who could take which exam and the like."

(female student, 20 years of age, author's translation)

From these student responses we can note that the cause of increased stress is in part the result of factors which are outside of the control of the faculty management (for example, isolation and the functioning of public transport). However, a part of the perceived stress clearly emerged as a result of the poor organization of classes, exam requirements, and the exams themselves, all of which could have been done differently.

Students of the Faculty of Agriculture, as the most numerous sub-group in the sample, issued fewer pieces of criticism aimed at decision-makers than the students from the other faculties, in terms of class organization, and especially in terms of exam requirements, and the exams themselves. This is an indication that this faculty was more responsive to the problems the students were facing during the state of emergency.

CONCLUSION

The presence of stress was noted among all the surveyed students of the UNS. Most of the respondents (64.36%) were moderately stressed (average overall score of 21.88). The number of those with an exceptionally high or exceptionally low level of perceived stress was relatively low, which is in accordance with what other researchers discovered studying stress among the student population in other countries.

A somewhat lower level of perceived stress was noted among the students of the Faculty of Agriculture (20.11) compared to that of the students attending other faculties (22.94).

Students of the Faculty of Agriculture were less critical of the decision-makers at their faculty regarding the organization of class and the conditions for the exam requirements and the exams themselves. As a result, the difference obtained in terms of the level of stress, at least in part, can be ascribed to the differences in class organization at a time of impeded function which resulted from the preventive measures that were put into place. Therefore, it is recommended that the management of the Faculty of Agriculture, in future, meet the demands of the student population to the extent that it is possible, but only when these demands are justified and conditioned by objective circumstances which the students have no influence over.

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VARIABILITY OF SOME MORPHOLOGICAL CHARACTERS IN SOLANUM TUBEROSUM L.

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SUMMARY

The experiments were carried out in order to evaluate morphologically the characteristics of some potato varieties (Desiree, Spectra and Arizona). These were also used as experimental variants, variant V1-Desiree, variant V2-Spectra and variant V3-Arizona. During the vegetation period from 20 to 100 days, the morphological development of Solanum tuberosum L plants was followed by establishing the length and formation roots, plant height, number of leaves/plant, leaf length, number of flowers/plant, number and weight of tubers.

Keywords: potato (*Solanum tuberosum* L.), plant height, number of leaves, tuber weight

INTRODUCTION

On the globe, the potato is spread in both hemispheres on all continents. After corn, the potato is the plant with the greatest ecological plasticity, being found in over 140 countries on six continents, where it is grown mainly for food purposes, but also for industrialization in order to obtain organic or mineral derivatives.

Solanum tuberosum the commonplace and well-known potato, present almost daily in one form or another on our table, has an ancient and glorious history commensurate with its nutritional qualities.

Although originally from America "found" by Europeans in 1492, it was only "discovered" in 1532, after the conquest of the state of Peru by the Spanish. The place of origin is known and undisputed, the sunny slopes and valleys of the Andes mountains and the surroundings of Titicaca in South America, but how it got to Europe and, especially who brought it, is a controversial issue. The oldest written attestation of the presence of this plants in Europe dates back to 1573 and is in the archives of a hospital in Seville, Spain.

Currently, the potato is cultivated in all continents, but especially in Europe where it owns about 60% of the world surface. The evolution of this culture in our country was not spectacular, the productions obtained were not at the level of other countries, of the pedoclimatic conditions and cultivated varieties. On the globe, the potato is spread in both hemispheres on all continents. After corn, the potato is the plant with the greatest ecological plasticity, being found in over 140 countries on six continents, where it is grown mainly for

food purposes, but also for industrialization in order to obtain organic or mineral derivatives. It was and still is the most authentic of field cultures.

MATERIAL AND METHOD

Observations were made based on biometric measurements of potato varieties (Desiree, Spectra, Arizona). The experimental variants were the V1-Desiree variant, the V2-Spectra variant and the V3-Arizona variant

At the beginning of the vegetation period (20) days, the morphological development of the *Solanum tuberosum* L plants was followed by establishing the length and formation of the roots, the height of the plant, the number of leaves/plant, the length of the leaves.

The morphological development of *Solanum tuberosum* L. plants (60) days after planting was achieved by determining the length and formation of roots, plant height, number of leaves/plant, leaf length.

At the end of the vegetation period (100) days after planting, the morphological development of *Solanum tuberosum* L plants was monitored by establishing the length and formation of roots, plant height, number of leaves/plant, length of leaves, number of flowers/plant, number and weight of tubers .

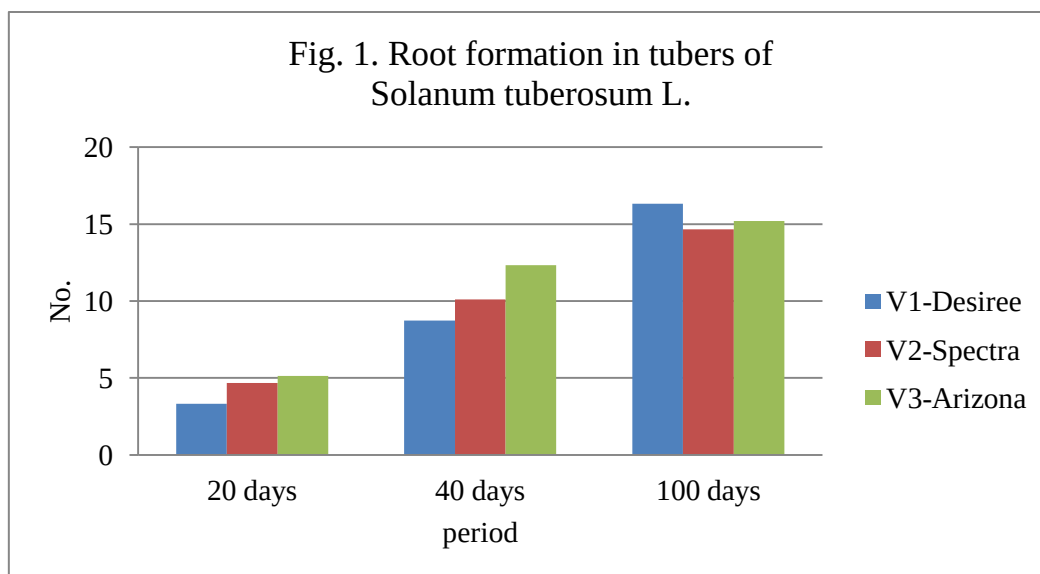
Monitoring during the observation period, biometric measurements were made of the length and formation of the roots, the height of the plant, the number of leaves/plant, the length of the leaves, the number of flowers/plant, the number and weight of the tubers. For our statistical interpretation we used 50 plants in the three observation periods and statically calculated the result.

RESULTS AND DISCUSSION

1. Root formation at *Solanum tuberosum* L. tubers.

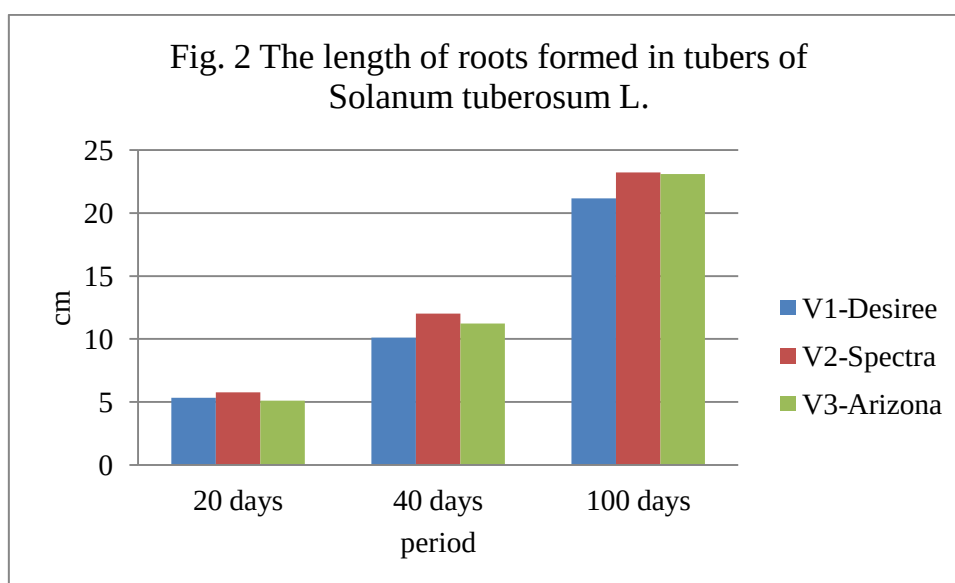
After 20 days from planting, root formation has a maximum value in the V3-Arizona variant with an average of 5.12 followed by the V2-Spectra variant with an average of 4.67. V1-Desiree having the minimum value of 3.33. Root formation in *Solanum tuberosum* tubers after 60 days from planting have different values from variety to variety. The V3-Arizona variant has the highest value with an average of 12.33 and the V1-Desiree variant has the minimum value of 8.74.

After 100 days from emergence, the varieties do not evolve much from each other, the V1-Desiree variant has a value of 16.33, the maximum value has the V3-Arizonan variant with an average of 15.21, and the minimum value has the V2-Spectra with an average of 14.67 (fig. 1).



2. The length of roots formed at tubers of *Solanum tuberosum* L.

After 20 days from planting, the root length has a maximum value for the V2-Spectra variant of 5.77cm, and close values for the V1-Desiree variant of 5.33 and the V3-Arizona variant of 5.11cm. The length of the roots after 60 days after planting the tubers is close to Variant V1, being the first in the ranking with 10.11cm tuber root length, followed by Variant V3-Arizona with 11.23 cm and Variant V2-Spectra with 12.02cm. The values of the length of the roots after 100 days from planting are close to the Spectra and Arizona varieties with an average of 23 centimeters, and the Desire variety has the smallest value of 21.17cm (fig. 2).

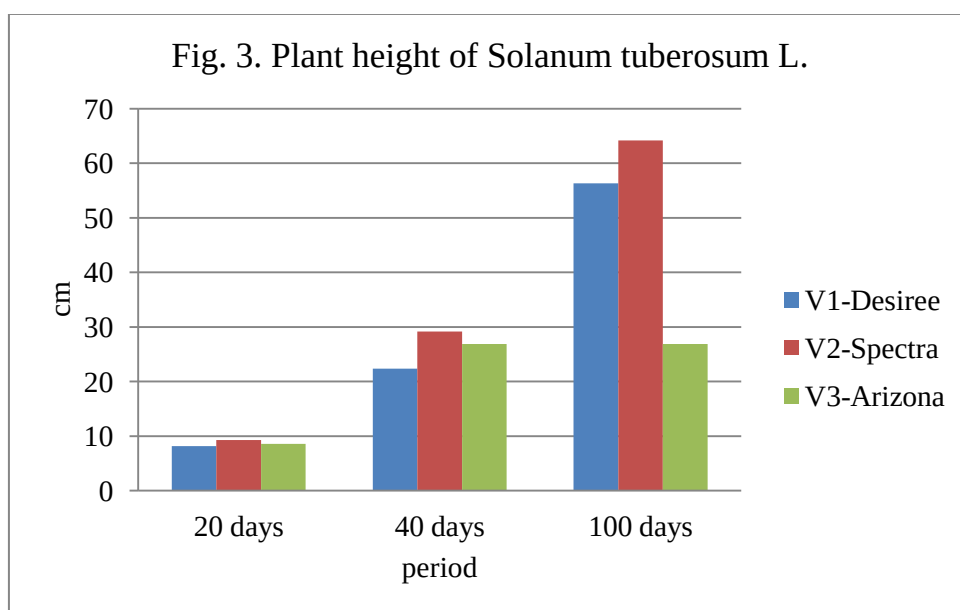


3. Plant height of *Solanum tuberosum* L.

At 20 days after planting, the height of the plants had a maximum value of 9.27 cm for the V2-Spectra variant, and for the V1-Desiree and V3-Arizona variants values close to 8.13 and 8.55, respectively.

After 60 days, the variants V2-Spectra and V3-Arizona have close plant height values of 29.15 cm and 26.88 cm, respectively, and for the variant V1-Desiree the plant height was 22.34 cm.

The height of the plants after 100 days showed the following minimum values for the V1-Desiree variant 56.33 cm and maximum values for the V2-Spectra variant 64.17 cm and 61.22 for the V3-Arizona variant (fig. 3).

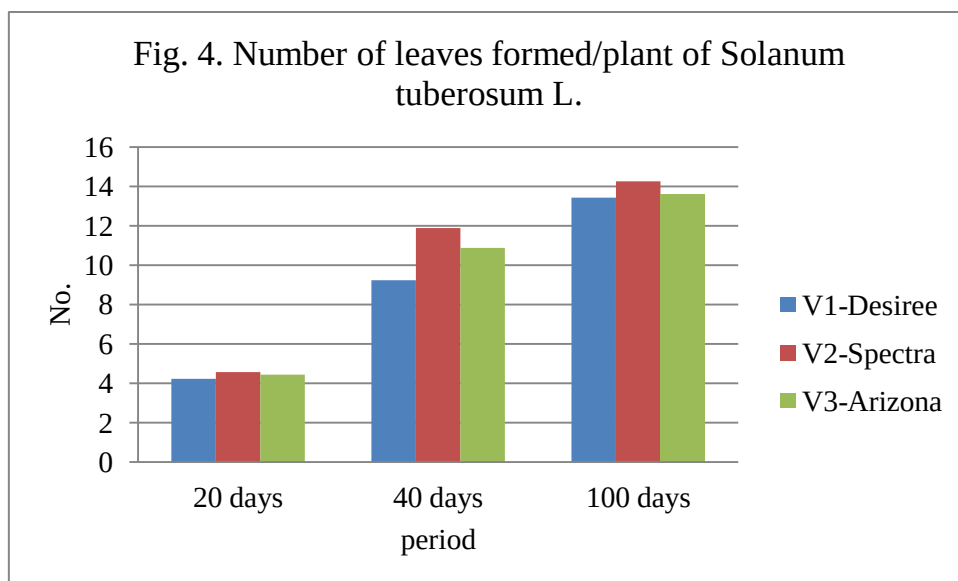


4. Number of leaves formed/plant of *Solanum tuberosum* L.

In figure 4 we can see that after 20 days after planting, the number of leaves differs very little between the 3 analyzed variants, a higher average number of leaves formed/plant of 4.56 has the variant V2-Spectra variant, then the variant V3-Arizona variant and the variant V1-Desiree with an average number of leaves of 4.44 and 4.33 respectively.

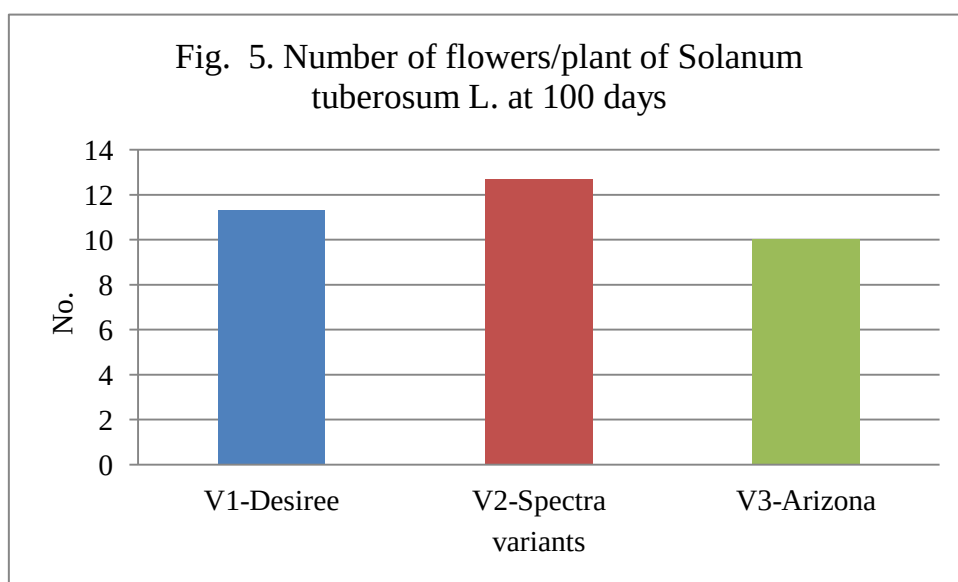
At 60 days after planting, the average number of leaves formed/plant did not differ much from variant to variant, respectively 9.23 leaves for variant V1-Desiree, 11.89 leaves variant V2-Spectra and 10.88 leaves at variant V3-Arizona.

After 100 days after planting, the number of leaves varied from a maximum value to the variant V2-Spectra with a number of 14.27 leaves/plant, followed by the V1 Desiree and V3-Arizona variants with a number of 13.44 and respectively 13.63 leaves/plant



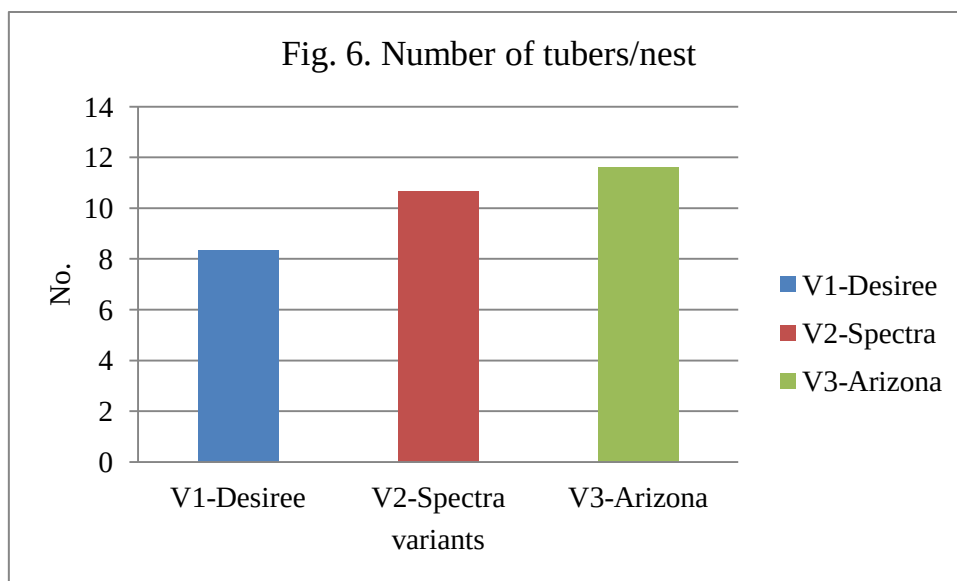
5. Number of flowers/plant of *Solanum tuberosum* L.

The number of flowers/plant of *Solanum tuberosum* is different from variant to variant. V2-Spectra has the maximum value of 11.67 and V3-Arizona the minimum value of 9. The V1-Desiree variant has a number of 9.33 flowers/plant being close to the V3-Arizona variant (fig. 5).

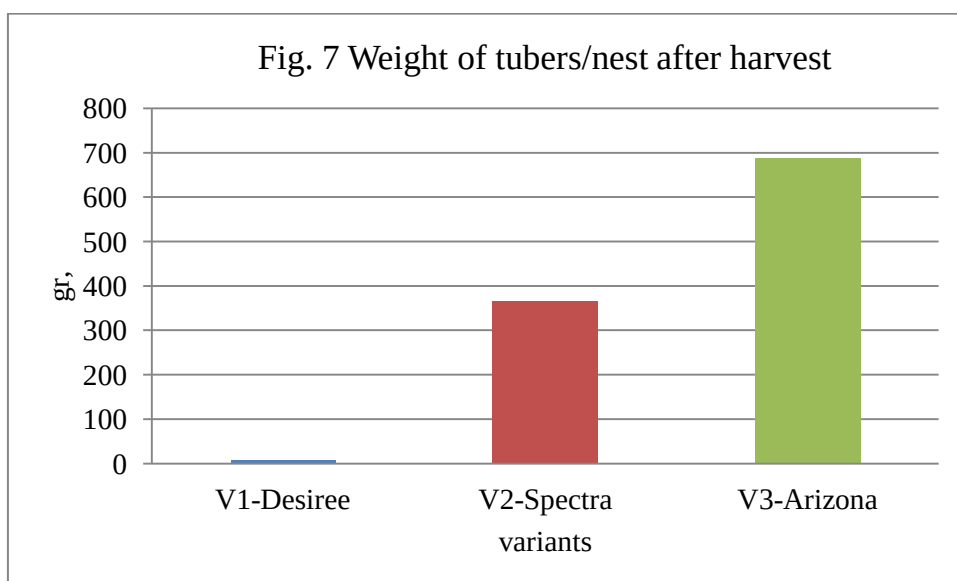


6. Number and weight of tubers/nest after harvesting *Solanum tuberosum* L. plants.

The number of tubers/nest is minimal in the V3-Arizona variety with an average of 10.33 tubers, followed by the V1-Desiree variety with an average of 8.33 tubers, and the maximum of 11.62 tubers belongs to the Spectra variety (fig. 6).



We can observe that the average weight of tubers/nest in the V3-Arizona variant is the highest with values of 687.50g. In the other two variants V1-Desiree and V2-Spectra, the average weight of tubers/nest is close, 389.75g and 364.25g respectively (fig. 7).



CONCLUSIONS

1. At the beginning of the vegetation period (20 days) root formation is different from species to species, with the V3-Arizona variant having the highest value. The length of the roots, unlike the previously mentioned character, is highlighted by the V2-Spectra variant.
2. After 60 days from emergence, root formation belongs to the same variant as at the beginning of the vegetation period, namely the V3-Arizona variant. As in the previous case,

the height of the plant belongs to the V2-Spectra variant, while the number of leaves belongs to the V3-Arizona variant.

3. The last vegetation period is the one between 100 days after planting with the following characteristics: root formation, plant height and number of leaves. These characters have distinctly different values from variety to variety, at the top of which is the V2-Spectra variant in terms of plant height and root formation. The number of leaves belongs to the V1-Desiree variant after 90 days from emergence, and to the V3-Arizona variant after 100 days from planting.

4. After harvesting, observations were made on: the number of tubers/nest, the number of eyes/tuber and the weight of tubers/nest. The characters, number of tubers/nest and weight of tubers/nest belongs to the variant V2-Spectra, and the number of eyes/tuber belongs to the variant V1-Desiree.

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WATER QUALITY MONITORING OF LUDAŠ LAKE

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SUMMARY

The aim of this paper was to show the results of the monitoring of some water quality parameters during late summer in 2022 in Lake Ludaš and compare them with the limit values for lakes in Serbian and Slovak legislation. Ludaš is a shallow lake in the province of Vojvodina in northern Serbia, near the city of Subotica. We analyzed nitrites and nitrates, total nitrogen and phosphorus, phosphates, ammonium, chloride, pH, conductivity using sensoring and photometric methods in water samples from the lake. According to both Serbian and Slovak regulations water quality of the lake can be classified as poor to bad.

Key words: pH, ammonia, chloride, phosphorus, water quality

INTRODUCTION

Ludaš lake is a shallow lake located in northern Serbia, near Subotica city. The quality of the available water intended for water supply depends on natural factors such as the geomorphological structure of the terrain, climatic conditions, recharge and drainage conditions, but also the influence of anthropogenic factors since a large part of the products of human activity reaches the water. Ludaš lake has been suffering significant water quality deteriorations for decades reflected in highly eutrophic conditions that could be easily observed as green-coloured water. The pollution originated from municipal and industrial wastewater, which had been discharged without treatment into Palić lake and later reached Ludaš lake by a canal dug to connect the two lakes when Palić lake dried up for cleaning. As a consequence, a large amount of mud accumulated, especially in the northern part, which increases the process of eutrophication and affects the degradation of the living world of the lake. In addition, Ludaš lake is surrounded by arable land from which, surplus nutrients after leaching, could reach it and the municipal waters of a few settlements that also contribute to

the pollution. (Seleši, 2006; Grabić et al., 2018). To control the water quality systematic monthly monitoring has been in place for at least two decades.

Monitoring presented in this paper aims to provide a more detailed overview of the changes in the water quality since we did water sampling weekly and were focused on the summer period when the worst conditions were expected for both the hydrological regime and weather parameters. In addition, we intended to interpret the result according to both Serbian and Slovak water quality regulations.

MATERIAL AND METHODS

The sampling within this research on Ludaš lake was done at three sampling points (Figures 1 and 2): Ludaš north, Ludaš middle and Ludaš south. The first location – Ludaš north, was next to the Visitor Center of the Public company Palić – Ludaš. This part is the widest part of the lake. The other two locations are situated at the parts of the lake that are much narrower. The second sampling site was located in the middle of the lake and was hidden by reeds. The last sampling site is located at the south of the lake next to the primary school in Šupljak village.

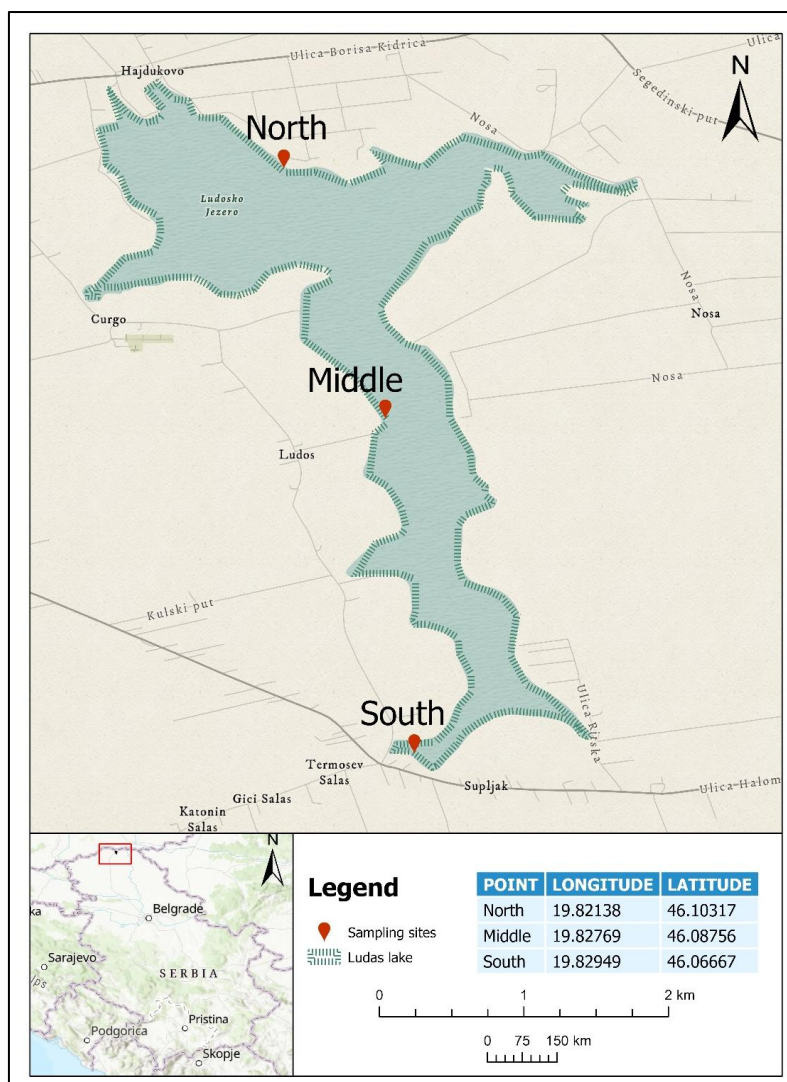


Figure 1. Locations of sampling sites at Ludaš lake

Sampling was done as a grab sample, by taking water directly from the surface 0 - 50 cm. Immediately after sampling water was frozen and kept in such a condition during transport and before analysing in the Laboratory for Waters, at the Faculty of Agriculture, University of Novi Sad. Immediately after defrosting basic water quality parameters were measured by sensing methods such as dissolved oxygen (DO), % of DO saturation, pH and conductivity. Suspended solids (SS) were analysed by the Photometer for suspended solids, Lovibond, Germany. Nutrient parameters i.e., nitrites - NO₂-N, nitrates - NO₃-N, orthophosphates - PO₄-P, total nitrogen - TN and total phosphorus - total P, were analyzed by the Multiparametric photometer, Lovibond, Germany.

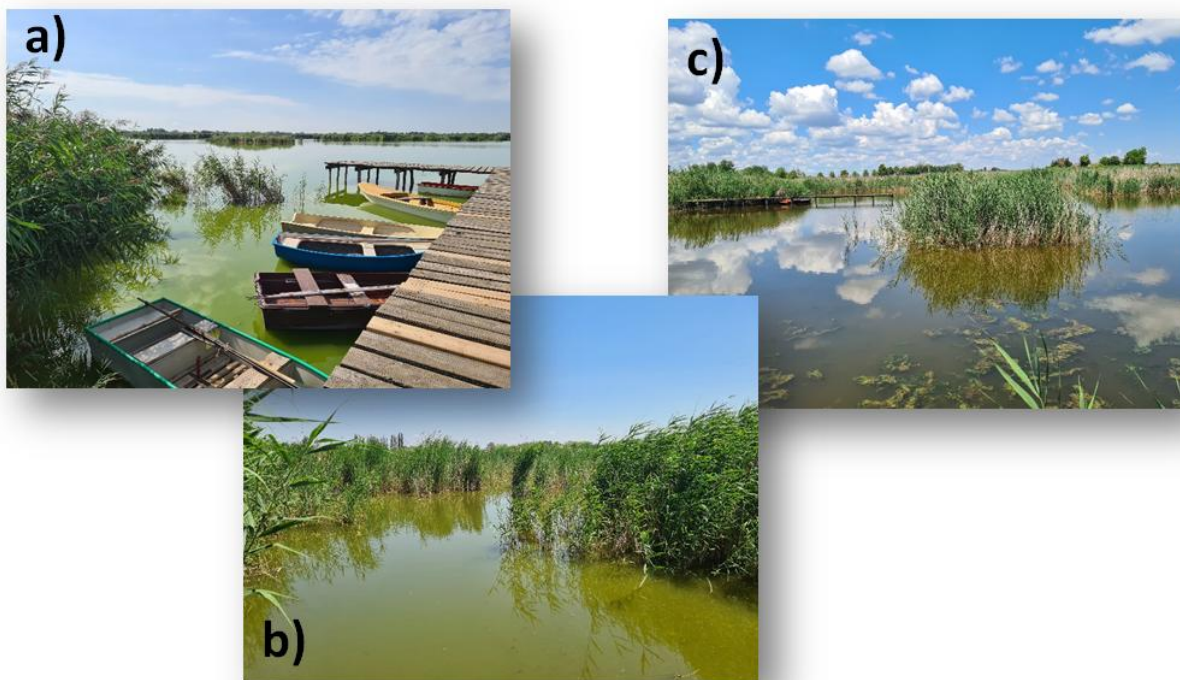


Figure 2. Photos of the sampling sites on Ludaš lake: Ludaš north; b) Ludaš middle; c) Ludaš south

Results were compared to the existing legislation on water quality, Serbian and Slovak, both in line with the Water Framework Directive of the European Union (Directive, 2000/60/EC).

RESULTS AND DISCUSSION

The results of laboratory analyzes of water samples from Lake ludaš are presented in Tables 1 and 2, in which they are subsequently compared with Serbian and Slovak water quality regulations.

Concerning Serbian regulation, there is the Regulation on the parameters of the ecological and chemical status of surface waters and the parameters of the chemical and quantitative status of groundwater (Regulation 74/2011/RS) which is in line with WFD (Directive, 2000/60/EC). The water quality of a certain type of waterbody is classified according to five classes set by the regulation, where blue colour designates excellent water quality (I class) and green refers to good water quality (II class) - both are acceptable. On a contrary, yellow, orange and red are grading water quality from moderate, poor to bad (II – V class), which is unacceptable (Table 1). In the case of Ludaš lake, the most unfavourable parameters are PO₄, TP and pH, while NH₄, NO₃ and Cl are mostly of desirable water quality. High conductivity is considered to be of natural origin, so the classification cannot be applied in this case.

Table 1. Water quality of Ludaš lake interpreted in accordance with Serbian water quality regulation Regulation 74/2011/RS, referring to lakes

Sample ID	PO4 (mg/lP)	NO2 (mg/lN)	Total P (mg/l)	NH4 (mg/l)	Total N	Susp. P.	NO3 (mg/l)	Chloride (mg/l)	PH	CONDUCTIVITY uS/cm
1.8. Ludaš 1	0,43	<0,01	0,75	0,19	7	91	<1	14,2	9,95	1257
1.8. Ludaš 2	0,47	<0,01	0,77	0,25	17,1	55	<1	14,8	9,23	1614
1.8. Ludaš 3	0,34	<0,01	0,54	0,08	9,2	41	<1	7,8	9,66	20500
7.8. Ludaš lake 1	<0,02	<0,01	0,6	0,15	7,9	-	<1	17,9	9,92	1277
7.8. Ludaš lake 2	0,22	<0,01	0,68	0,11	9,6	-	<1	14,2	9,20	1715
7.8. Ludaš lake 3	0,31	<0,01	0,41	0,16	12	-	<1	10,4	9,40	19800
14.8. Ludaš lake 3	0,23	<0,01	0,54	0,23	11,5	-	<1	12,7	9,91	2290
14.8. Ludaš lake 1	0,33	<0,01	0,38	0,06	5,4	-	<1	12,1	10,21	1345
14.8. Ludaš lake 2	0,1	<0,01	0,44	0,46	8,2	-	<1	11,8	9,23	1748
23.8. Ludaš sever 1	0,54	<0,01	0,54	0,19	7,3	76	<1	8,2	9,90	1008
23.8. Ludaš sever 2	0,45	<0,01	0,17	-	-	51	<1	20,4	9,94	869
31.8. Ludaš 1	0,47	<0,01	0,45	0,1	5,1	69	<1	13,6	9,89	1252
31.8. Ludaš 2	0,8	<0,01	0,97	0,31	15,9	107	<1	6,7	9,56	1521
31.8. Ludaš 3	0,34	<0,01	0,97	0,18	19,9	107	<1	6,9	9,58	22900

According to limit values, water quality is divided to 5 quality classes by color, blue (I.-II.) > green (II.-III.) > yellow (III.-IV.) > orange (IV.-V.) > red (over limit of quality class V.). Limit values for conductivity, suspended particles, nitrites and total nitrogen are not found in this regulation for lakes with altitude to 200m asl.

Surface water regulations in the Slovak Republic are based on the implementation of European legally binding documents, i.e. WFD (Directive, 2000/60/EC). In Slovak legislation, water quality grades were abolished and combined into one value (Table 2). Green cells mean good quality status and red cells mean over limit. The last row shows limits of this regulation. In Slovakia there are only general regulations for surface water and for drinking water supply sources (Regulation 269/2010), which includes all types of water sources such as rivers, lakes and reservoirs. There is no general regulations for suspended particles only suspended particles dried at 105 °C with limit mg/l 900 and suspended particles annealed at 550 °C with limit mg/l 640. So we could not compare suspended particles with this regulation limits.

Similary to Serbian regulation water quality parameters measured at sampling sites on Ludaš lake which are unfavourable are PO4, TP and pH.

Table 2. Water quality of Ludaš lake interpreted in accordance with Slovak water quality regulations for surface water (Regulation 269/2010, annex 1)

Sample ID	PO4 (mg/lP)	NO2 (mg/lN)	Total P (mg/l)	NH4 (mg/l)	Tot al N	Sus p. P.	NO3 (mg/l)	Chlori de (mg/l)	PH	CONDUCTI VITY uS/cm
1.8. Ludaš 1	0,43	<0,01	0,75	0,19	7	91	<1	14,2	9,95	1257
1.8. Ludaš 2	0,47	<0,01	0,77	0,25	17, 1	55	<1	14,8	9,23	1614
1.8. Ludaš 3	0,34	<0,01	0,54	0,08	9,2	41	<1	7,8	9,66	20500
7.8. Ludaš lake 1	<0,02	<0,01	0,6	0,15	7,9	-	<1	17,9	9,92	1277
7.8. Ludaš lake 2	0,22	<0,01	0,68	0,11	9,6	-	<1	14,2	9,20	1715
7.8. Ludaš lake 3	0,31	<0,01	0,41	0,16	12	-	<1	10,4	9,40	19800
14.8. Ludaš lake 3	0,23	<0,01	0,54	0,23	11, 5	-	<1	12,7	9,91	2290
14.8. Ludaš lake 1	0,33	<0,01	0,38	0,06	5,4	-	<1	12,1	10,2 1	1345
14.8. Ludaš lake 2	0,1	<0,01	0,44	0,46	8,2	-	<1	11,8	9,23	1748
23.8. Ludaš sever 1	0,54	<0,01	0,54	0,19	7,3	76	<1	8,2	9,90	1008

23.8. Ludaš sever 2	0,45	<0,01	0,17	-	-	51	<1	20,4	9,94	869
31.8. Ludaš 1	0,47	<0,01	0,45	0,1	5,1	69	<1	13,6	9,89	1252
31.8. Ludaš 2	0,8	<0,01	0,97	0,31	15,9	107	<1	6,7	9,56	1521
31.8. Ludaš 3	0,34	<0,01	0,97	0,18	19,9	107	<1	6,9	9,58	22900
limits	0,2-0,4	0,02	0,4	1	9		5	200	6-8,5	1100

Some values such as pH or conductivity could be included in a bad quality class because the measurement was not performed directly in the lake, but was made from a sample that was taken a few weeks or a month ago and frozen, which could affect the resulting values. When we compare both regulations, one can see that both are presenting similarly unfavorable conditions in the lake. The difference is only that Serbian regulation is providing more details by classes. In Slovak regulation, there are limits in classes for drinking water supply sources like reservoirs; and applying to Ludaš lake for all sampling dates during August, the water quality did not change much.

CONCLUSION

When we compare the Slovak regulation influenced by the EU and Serbian, we can see that in Serbian regulation there are quality classes according to different limit values, but in Slovak, there is only one limit value. Slovak regulation is done for surface water as a whole, and Serbian for lakes and rivers separately. Regarding the water quality in Ludaš lake, according to Serbian regulation, we can say that the quality ranges from classes I to V, but 4 parameters did not have limit values. In general, it can be said that the quality in the lake is not good, high pH and phosphorus indicate eutrophication. According to Slovak regulations, water quality is bad as more parameters exceed the limit values - especially pH, conductivity and total P.

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INDOOR FOOD PRODUCTION IN ORNAMENTAL AQUARIUM-BASED AQUAPONICS SYSTEM AS A CIRCULAR ECONOMY APPROACH

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SUMMARY

The combination of aquaculture and hydroponics technology represents a sustainable synergistic system called aquaponics (AP). The system simulates natural processes between three different groups of organisms – fish, bacteria, and plants. The AP converts toxic substances produced as a fish by-product into nutrients needed by plants, saves space, does not depend on climatic conditions, provides organic food because no chemical preparations and pesticides are used, and thus prevents environmental pollution. It does not require soil as a resource for plant growth and saves significant amounts of water due to water circulation in the system. Aquaponics is in use for many years in the world but still is in the development phase in Serbia. Thus, the project presented here aimed to create small-scale, ornamental aquarium-based aquaponics to familiarize the interdisciplinary team as well as to teach students and all interested parties what processes aquaponics includes and how to implement the system.

Key words: Media-bed aquaponics, Circular economy, Nutrient cycle, Sustainability

INTRODUCTION

With the increase in the world's population, it is necessary to produce as much food as efficiently as possible, especially in difficult conditions, caused by frequent drought events, and unfavorable climatic conditions in general. To feed a population of 9.6 billion people in 2050, there is a need to produce 70% more food than today (Pereira et al., 2020). Agricultural production is expected to increase by 15% over the next decade, even though agricultural land will remain the same (OECD/FAO, 2019). This means high pressure on environment, especially on soil to produce food. Agricultural intensification is already degrading soils and reducing their long-term capacity to produce food. This is especially important because about 99% of our food is produced on soil (Kopittke et al., 2019). Also, the pressure is high with regard to nutrient pollution, overuse and improper use of herbicides, pesticides, fertilizers, etc. In addition, conventional agricultural production under irrigation

is the number one consumer of water, about 70% of the global freshwater consumption (Li et al., 2022). As reported by the UN (<http://www.unwater.org>), water scarcity already affects around 1.2 billion people while 500 million people are approaching this situation where the frequency and intensity of droughts have increased. To mitigate the negative implications of conventional methods, alternative food production systems have been developed (Breitenstein & Hicks, 2022). To overcome the issues, agriculture-based sustainable farming was suggested to produce food by conserving water and recycling nutrients, and converting wastes (water) into high-value resources (Wongkiew et al., 2017). Moving towards a more circular economy could deliver benefits (Avellán et al., 2022; Kampelmann et al., 2021). The concept of *circular economy* is defined as “a production and consumption model, which involves sharing, renting, reusing, repairing, renovating and recycling existing materials and products for as long as possible” (European Parliament, 2015). The circular economy is “to bring about perfect circles of slow material flows, to prompt a shift from consumer to user, and to enable a decoupling of resource use and environmental impact from economic growth” (Lazarevic & Valve, 2017). Sustainable production of quality food under premises of circular economy reducing the flow of resources (nutrients) out of the system is crucial for ensuring food security while maintaining non-renewable resources such as phosphorus and reducing environmental impacts of agricultural practices (Jurgilevich et al., 2016). Aquaponics is emerging as an alternative for sustainable agriculture which meets the philosophy of circular economy, decreasing not only the need of external resources but also increasing their use efficiency (Fernández-Cabanás et al., 2022). Aquaponics (AP) is a sustainable synergistic system that combines aquaculture and hydroponics technology (Baganz et al., 2021). Aquaculture according to FAO (1988) is “*the farming of aquatic organisms, including fish, molluscs, crustaceans and aquatic plants*” (<https://www.fao.org/>). Hydroponics is often called “cultivation in water”, or “cultivation without soil” (Sorenson & Relf, 2009). The pith is that instead of soil, a plant is taking nutrients from the water. The symbiosis between these two systems, aquatic species and plants, is reflected through the microbial transformation of aquaculture effluents (toxic substances for fishes) that plants use as a nutrient source, and thus the water going back to the aquaculture sub-system purified (Yep & Zheng, 2019). Also, nutrient removal depends on a growing media that filters ammonia-rich fish waste and solid waste (Hamid et al., 2022). Hydroponics helps to mitigate some environmental and climate problems that are usual to conventional farming: helps to alleviate some of the pests, diseases, and environmental issues such as heavy rain, drought, and soil nutrient problems that are common to modern crop farming (Hemathilake & Gunathilake, 2022). Hydroponic (HP) systems are generally the production of plant/vegetable crops in areas smaller than that required under field production (El-Sayed, 2006). There are numerous benefits of an HP system. It does not require soil as a resource for plant growth and saves significant amounts of water due to water circulation in the system, saves space, does not depend on climatic conditions, provides organic food because no chemical preparations and pesticides are used, and thus prevents environmental pollution. The AP is still in the development phase in Serbia, although it has been used in the world for many years. To get more familiar with the concept of circular economy, the process of simulating natural processes in the aquatic environment between three different groups of

organisms (fish, bacteria, plants), and the phases of developing aquaponics solutions (conceptual model, physical model), the interdisciplinary team from the Faculty of Agriculture, e.g. the group of students of Water Management, Horticulture and Landscape Architecture and Vegetable and Crop Production have designed the AP model.

MATERIAL AND METHODS

The AP model is implemented as a media-bed flood-and-drain system that consists of several subsystems: an ornamental aquarium with fishes; a growing bed with Light Expanded Clay Aggregate (LECA); and a system of pipes and pumps, that connects the previous two by bell-siphon.

Every AP system requires fish feed and fresh water, with the main product being the harvested fish. The by-products are wastewater and sludge which produce harvestable vegetables along with wastewater. In HP, the inputs are water and fertilizer, and in AP the two systems are combined, using the wastewater from the fish to fertilize the plants, and the “treated” water from the plants flows back to the fish. Additionally, both can be harvested fish and vegetables, within the same system. This represents the standard AP system. In this paper, we represent an ornamental aquarium-based AP system, where the aquaculture system has an ornamental purpose among others, and the produced food can be used.

A population of fish lives in the fish tank – in this case - the ornamental aquarium. Fish are fed with standard commercial food, and the fish excretes the remains of digested food. Also, some food may remain uneaten. Thus, the water in the aquarium/tank contains more and more fine particles and soluble compounds. Some of these can be toxic to fish, especially ammonia. From the aquarium, the water flows to the media bed which is planted with chosen plants. We have used LECA substrate which is *Light Expanded Clay Aggregate* and is a well-known substrate for HP. Besides that purpose, LECA also acts as a biofilter. LECA globules are porous and have a huge surface internal area. They provide plenty of space for microorganisms, mostly bacteria. Some of these bacteria transform the toxic ammonia, produced by fish into nitrate. Nitrate and ammonia are both taken up by plants. Thus, the water is suitable for fish again. After that, the water goes to the fish tank. This closes the cycle. Basically, every system, classified as AP, has the following components: a fish tank, a solids removal unit which can be either a settler or a drum filter or both, a biofilter unit which can be LECA or some other substrates with a high surface area for bacteria to grown on, a plant growing unit, which can take different forms, and pumps, pipes and a sump for the water exchange. These components can be combined. Alternatively, each component can be a separate unit.

The choice of AP system you want to use depends heavily on the objectives. For example, do you want an AP system for commercial crop production, household sufficiency, education, or for greening and decoration, etc? These are the questions you have to be able to answer clearly before you begin your AP system.

RESULTS AND DISCUSSION

The AP model is installed at the Faculty of Agriculture (UNS). The system is composed of an 80x45x45 cm aquarium with seven goldfishes (*Carassius auratus*) and three *Coridoras paleatus* which are fed once per day with commercial food. The bottom is 2 cm filled with gravel (1-3mm).

A trough for growing plants (80x80 cm) is filled with LECA (around 7 cm deep). A system of pipes and pumps connects the previous two by a bell-siphon. The water flows for 15 min. every three hours. Also, air stones connected to an air blower were installed in the aquarium to supply oxygen to the fish. The coarse sponge filter is placed in the aquarium, at the end of the pipe that drains the water from the trough, as well as around the siphon in the growing bed. The rest is naturally purified by the system. In the first phase, mint (*Mentha spicata*) and bamboo (*Equisetum japonicum*) were planted. In the next phases, it is planned to grow garlic, lettuce, and herbs.

In the first step (Figure 1), an aquarium was installed with water into which a pH stabilizer was added, to bring tap water to a suitable state, i.e. to prepare it to be a favorable habitat for the fish, which were added after some time. After setting up the tank, the preparation process included running it for a few days, adding a conditioner that removes harmful substances (chlorine and heavy metals) and thus protects the fish. In the second step, the water quality was checked, where the pH value, electrical conductivity, temperature, alkalinity, total chlorine, nitrites, nitrates, and total hardness were determined.



Figure 1. Aquarium preparation process

In the third step, an area for growing plants, i.e. a trough, was prepared, where it was necessary to attach siphons to which hose was connected to the submersible pump installed in the aquarium (Figure 2). Then, the trough was filled with a substrate for growing plants (LECA), which previously had to be washed off until it was completely clean because otherwise, it could cause pollution of the aquarium water (Figure 3). The drainage part, the siphon, and the discharge hose are covered with a cotton filter, to additionally ensure the entry of impurities from the substrate.

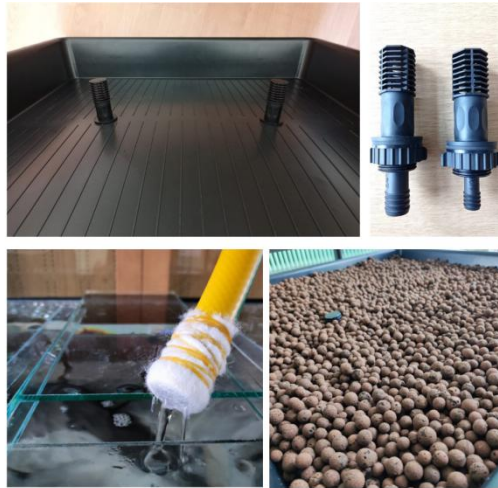


Figure 2. The process of preparing a media-bed



Figure 3. Preparation of the substrate for plants

The fourth step involved planting selected plants. In the fifth step, the flooding and drainage time (on which the system is based) is set using a mechanical timer. It is set so that every three hours water flows through the system for 15 minutes, in order to provide the plants with a sufficient amount of nutrients from the water. At the same time, the roots of the plants should not be damaged due to standing in the water for too long, and also should not be under the stress and thus slowing down growth and development. Figure 4 shows the installed AP model.



Figure 4. The AP system

CONCLUSION

The Media Bed method is one of the most popular choices because the design is relatively simple, which makes it easier to manage. When choosing the grow media it is important to analyze many criteria. For this purpose, LECA was chosen, among others, because of its drainage, water retention, aeration, and insulative properties. Some benefits of media bed aquaponics (among other methods) are: minimal costs, support wide variety of plants, highly customizable, etc. The model of AP can help students to a deeper understanding of ecosystems and causal patterns because of simulating natural relationships and processes in the aquatic environment between three different groups of organisms. Also, the development of environmental awareness due to reduced environmental risks, saving water, circular economy, etc., could be achieved. As Milliken et al. (2022) in their work stressed: “aquaponics promotes ecological literacy among students, thereby enabling future professionals of various careers whose activities are affected by - and have consequences for environmental issues, and provides a pathway for introducing the concepts of sustainable development and the circular economy to higher education curricula”. The reason lies in the fact that “as a nature-based solution addresses many socio-environmental challenges, including food and water security, water pollution, human health, and climate change as a balanced closed-loop system”

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APPLICATION OF SPI (STANDARDIZED PRECIPITATION INDEX) FOR IDENTIFICATION OF DRY AND WET PERIODS

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SUMMARY

Precipitation is one of the most important factors in the Earth's water cycle. The precipitation regime is affected by climate change, which represents one of the most significant problems facing the whole world. Climate change is a problem that was noticed not so long ago on the territory of our country and it greatly affects the agricultural activities. In the area of Vojvodina, one of the main tasks of water management is dealing with excess water which occurs as a result of extreme rainfall and removing it from the land and the ground. Vojvodina is highly endangered by climate change and climate extremes are expected to become more frequent. SPI has become the most commonly used index in the practice of drought monitoring, but it can also be used for the analysis of wet periods. Therefore, the aim of this research was to show the possibility of using SPI for both cases. The climatic characteristics of Vojvodina were analyzed for the period from 1971 to 2017 on the basis of meteorological data of eight meteorological stations of the Republic Hydrometeorological Institute of Serbia. There are seven stations in Vojvodina (Palić, Kikinda, Sombor, Zrenjanin, Rimski Šančevi, Vršac and Sremska Mitrovica), and the eighth station is the meteorological station Belgrade. By comparing the maps for each month, it can be concluded that during the observed period (June-September, 2010) very wet conditions occurred throughout the territory of Vojvodina. Also, for the observed period (June-September, 2017) it can be concluded that droughts of different duration and intensity occur. By mapping the SPI, the necessary bases can be obtained for the analysis and assessment of the risk of hydrometeorological extremes in order to mitigate unwanted consequences.

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